

Correct Predictions, Wrong Reasons: On the Limits of Metrics for Measuring Reasoning Ability

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Prediction metrics such as accuracy, F1, MCC, AUROC, and paired or group-based success rates are often interpreted as measures of a model's reasoning ability. This paper establishes a validity boundary for that interpretation. We show that finite output-based metrics are subject to three general limitations: mechanism non-identifiability, because identical benchmark outputs can arise from different mechanisms; metric non-monotonicity, because a shortcut model can outscore an imperfect semantic model; and non-separability, because an observed controlled-success rate does not uniquely determine its semantic and residual non-semantic components.

We examine the empirical consequences through a secondary analysis of two frozen matched-evaluation archives and a raw-prediction group analysis covering 3-SAT, 2-SAT, and graph coloring. The analysis compares conventional and controlled metrics with checkable positive-side witnesses, quantifies model-ordering discordance, and stress-tests controlled conclusions under pair rematching, confidence-bound choice, reporting thresholds, and shortcut-ceiling specification. At the precision reported in the source tables, accuracy reverses the witness-based ordering in 3/16, 10/115, and 25/62 strict same-size model comparisons on 3-SAT, 2-SAT, and graph coloring, respectively. A Clopper–Pearson lower bound on matched two-sided separability still reverses the witness ordering in 3/16, 9/116, and 27/63 comparisons. Pair rematching changes ADR by as much as 0.364, and reported capability boundaries move under both threshold and shortcut-account choices. Across 130 constraint model–size slices, no evaluable group reaches ten successful matched pairs in 88 slices, showing that isolated pair correctness often does not persist across controlled variants.

These findings do not make metrics useless. They show that conventional and controlled metrics support claims about observable behavior under a stated benchmark and design, not an unqualified ordering of latent reasoning ability. Reasoning-oriented evaluation should therefore report metrics together with witnesses, controlled interventions, nuisance-factor audits, construction sensitivity, and claim-specific validity arguments.

Additional Key Words and Phrases: reasoning evaluation, metric validity, matched evaluation, witnesses, shortcut learning

1 Introduction

Machine-learning and large-language-model (LLM) systems are increasingly used for software-engineering tasks that are often described as requiring reasoning, including vulnerability detection [7, 36], constraint solving [1, 42], program-behavior analysis [12], test generation [18], and debugging [38]. Reasoning ability is critical for such systems because it affects deployment, diagnosis, and benchmark design. A model whose predictions are grounded in the semantic evidence that should determine the answer is more likely to be trustworthy, reliable, and transferable. In contrast, a model that relies on code style, formula encoding, project artifacts, API names, or dataset-construction artifacts may perform well on a benchmark but fail when those correlations change. Similarly, a model that fails because it lacks semantic reasoning requires different improvements from one that merely needs threshold tuning or calibration.

The performance of such systems is commonly summarized using output-based metrics. In prediction tasks, these include accuracy, precision, recall, F1, MCC [4, 21], Cohen's Kappa [6], G-mean, Youden's J , Wilson lower confidence bounds, AUROC, and related classification or ranking metrics. In generative code-reasoning tasks, they may include exact-match or pass@ k metrics. In test generation, they may include task-specific measures such as line or branch coverage. These metrics are useful and often necessary. They quantify observable performance on a benchmark

distribution. However, a higher metric value does not necessarily indicate stronger reasoning ability.

This paper argues that no metric computed only from a finite sequence of ground-truth labels together with model predictions or scalar scores can, in general, serve as a measure of reasoning ability. Such metrics measure agreement, ranking quality, or another observable property of benchmark behavior. A higher value may reflect stronger semantic reasoning, but it may also arise because non-semantic cues are easier to exploit on the benchmark distribution. Consequently, a higher metric score does not necessarily imply stronger reasoning or a better model for the intended task.

Existing and proposed metrics. Recent surveys indicate that conventional prediction metrics remain the dominant way to measure and compare ML-based software-engineering systems [15, 28, 29, 35, 39, 43]. This practice is not unique to software engineering. Broader surveys of deep learning also show that accuracy, precision, recall, F1, ROC/AUC, error metrics, ranking metrics, and task-specific output metrics are widely used across regression, classification, computer vision, natural-language processing, and related areas [32]. These metrics address important statistical and experimental concerns. MCC summarizes all four entries of the confusion matrix and is often preferred in imbalanced binary classification [4, 21]. Cohen’s Kappa adjusts observed agreement by subtracting the agreement expected from the marginal class and prediction distributions [6]. G-mean and Youden’s J emphasize balanced sensitivity and specificity [13, 41]. Wilson lower confidence bounds provide conservative estimates of observed success rates under sampling uncertainty [37]. AUROC avoids committing to a single threshold by measuring how well scores rank positive examples above negative examples [8].

Recent controlled metrics address a different weakness by changing the evaluation design rather than only changing the metric formula. In vulnerability detection, Ding et al. [7] showed that code language models can perform much worse under realistic evaluation settings than standard benchmark results suggest. They introduced pair-oriented metrics that test whether models distinguish similar programs with different vulnerability labels. In satisfiability solving, Zhang et al. [42] introduced Accurate Differentiation Ratio (ADR), which evaluates whether an LLM distinguishes matched SAT/UNSAT formulas rather than merely achieving high marginal accuracy. These metrics are valuable because they make some shortcut strategies less useful. A higher controlled-metric score therefore provides stronger evidence under the chosen evaluation design, but it should not be interpreted as a direct ordering of reasoning ability.

Formal argument. We develop the central claim using a simple distinction between semantic and non-semantic information in a benchmark instance. Semantic information is the information that should determine the ground-truth label under the intended task semantics. Non-semantic information should not determine the label, but it may still correlate with the label or affect a model’s prediction. We use the term side-channel cue for non-semantic information that becomes predictive for a particular model or benchmark.

The distinction is concrete in our target domains. In vulnerability detection, semantic information includes the program relationships that make a vulnerability present or absent, such as data-flow and control-flow relationships between sources, checks, and sinks. In SAT solving, semantic information includes the variables, literals, negations, clauses, and connectives that determine satisfiability. In both domains, non-semantic information may include presentation choices, naming conventions, project-specific or generator-specific artifacts, and other features that do not determine the label by the intended task semantics. Such features need not be harmful in every benchmark, but when they correlate with labels or influence model predictions, they can support shortcut-based success.

This semantic/non-semantic gap leads to two general limitations for prediction-output metrics. First, mechanism non-identifiability: any metric computed from a finite vector of labels and predictions cannot distinguish two models that make the same predictions on the benchmark, even if one relies on semantic reasoning and the other relies on shortcut cues. Second, metric non-monotonicity: even when two models make different predictions, a higher metric score does not necessarily indicate stronger reasoning. Semantic reasoning may be difficult, and a model that attempts to use semantic evidence may still fail to capture the full relationship between instances and labels. A shortcut-based model can therefore receive a higher score when the benchmark contains non-semantic cues that are easier to exploit and more predictive on that benchmark distribution.

The same limitations extend to score-based metrics such as AUROC. Scores and rankings avoid the information loss caused by thresholding a score into a binary prediction. Nevertheless, a higher AUROC indicates a more favorable benchmark ranking, not necessarily stronger reasoning, because a non-semantic cue may separate the benchmark classes more effectively than an imperfect semantic model.

Controlled metrics [7, 42] introduce a third limitation that we call non-separability. Pair, group, and cluster metrics construct more demanding evaluation units, but their observed success probabilities still mix semantic success with residual shortcut or guessing success. For example, group metrics can attenuate weak independent shortcut effects by requiring repeated success across related instances. However, this benefit depends on how the controlled instances are aggregated. An aggregation rule that works for one dataset, group construction, model family, or problem domain need not transfer to another. Thus, controlled metrics can move measured success closer to semantic success in a particular setting, but they do not separate the semantic component from the residual non-semantic component.

Group evaluation also tests a distinct property that instance- and pair-level averages discard: stability across controlled variation. The intuition is the same as recognizing a dog after a rotation or crop that preserves its defining content. If a model tracks the task-relevant structure, its judgment should survive incidental changes and should change when a controlled edit changes the correct answer. In our constraint tasks, a group contains repeated matched positive–negative pairs around a common construction. Consistent success across that neighborhood is therefore stronger behavioral evidence than an isolated correct pair. It is still not equivalent to reasoning: a cue shared across the group can produce consistent labels, and a label-only group score does not check a satisfying assignment or valid coloring.

Model-level instantiations. We instantiate this argument for common model families. For attention-based models, token representations are combined through attention weights [34]. The final score can aggregate contributions from both semantic and non-semantic tokens. For graph neural networks and other message-passing models, program graphs are transformed by aggregating information over graph neighborhoods [16]. A vulnerability detector may therefore perform well by capturing a vulnerability-relevant data/control-flow slice, or by exploiting local features such as API names, identifier names, comments, or graph patterns. These instantiations illustrate how a model that exploits easier non-semantic regularities can equal or exceed the benchmark score of a model that more faithfully captures the intended semantics. This perspective is consistent with broader observations from shortcut learning. Models may learn decision rules that perform well on standard benchmarks but fail under more challenging or shifted test conditions [10].

Empirical argument. We examine the practical consequences in constraint reasoning with LLMs. The frozen archives contain matched 3-SAT and 2-SAT formulas together with graph-coloring instances. Satisfiable formulas and colorable graphs admit inexpensive, checkable positive witnesses:

a satisfying assignment or a valid coloring. These witnesses are stronger observable evidence than a correct category label, although they remain asymmetric because the negative side would require an UNSAT or non-colorability proof. We compare conventional metrics, ADR, matched two-sided separability, confidence-bounded variants, and witness-validated success. We also quantify model-ordering reversals and the sensitivity of controlled conclusions to pair matching, thresholds, confidence bounds, and shortcut-ceiling assumptions.

The experiments do not “validate” Propositions 1–3; those propositions follow from their definitions. Instead, they test whether the failure modes identified by the theory occur in realistic evaluation pipelines and whether common remedies remove them. The answer is mixed: witnesses and controlled designs reveal failures hidden by marginal metrics, but the resulting measurements remain design dependent and do not identify a unique reasoning mechanism.

Implications. Our analysis does not imply that prediction metrics are useless. Rather, it clarifies their validity boundary. Accuracy, precision, recall, F1, MCC, Kappa, G-mean, Youden’s J , Wilson bounds, AUROC, ADR, group-based ADR, pass@ k , and coverage metrics summarize different observable properties of model behavior. They address class imbalance, threshold dependence, marginal chance agreement, sampling uncertainty, pairwise differentiation, group-level consistency, accepted generation, or structural reachability. These summaries are often useful and necessary, but their values order models by the measured behavioral criterion, not necessarily by reasoning ability.

Controlled metrics should therefore be understood as metrics over more informative evaluation designs, not as direct measures of reasoning ability. ADR, group ADR, and cluster metrics can expose failures that ordinary accuracy or F1 may hide because they evaluate matched pairs, repeated success, or related variants. A higher controlled score indicates better performance under that design, but it need not indicate stronger reasoning because residual shortcut success and the aggregation rule can still affect the ordering.

Reasoning-oriented evaluation should therefore combine metrics with controlled interventions and additional evidence. Controlled metrics such as ADR, group ADR, and cluster metrics summarize behavior on matched pairs, related variants, or vulnerable-to-fixed patch directions. These designs test semantic sensitivity and make a high score more informative, but they do not by themselves establish a monotone relationship between metric value and reasoning ability. Complementary perturbations can change non-semantic factors while preserving semantics to test invariance. When available, checkable witnesses should be used to validate semantic success. When witnesses are unavailable or difficult to validate, metamorphic relations [3] and behavioral perturbation tests [26] provide weaker but still useful evidence. Metrics summarize the outcomes of these interventions; they do not certify reasoning ability.

This paper makes the following contributions:

- We formalize why prediction and score-based metrics are not, in general, reliable measures of reasoning ability. Using a distinction between semantic and non-semantic information, we establish mechanism non-identifiability and metric non-monotonicity: models with different reasoning behavior can receive the same metric value, and a shortcut-oriented model can receive a higher value than a model that captures the intended semantic relationship imperfectly. The analysis covers confusion-matrix metrics and score-based metrics such as AUROC.
- We analyze controlled pair, group, and cluster metrics, including ADR and its group-level extensions. These designs can reduce the usefulness of some non-semantic cues, but their observed success still combines semantic success with residual shortcut or guessing success. We formalize this non-separability and characterize the calibration tradeoff introduced by

equal-group versus equal-pair weighting, variance-sensitive aggregation, repeated-success thresholds, dependence among instances, and benchmark construction.

- We conduct a source-separated secondary analysis of matched constraint-reasoning experiments. The analysis covers 123 model-size rows across 3-SAT, 2-SAT, and graph coloring and compares metric levels and model orderings with checkable positive-side witnesses. A separate raw-prediction analysis covers 130 model-size slices across the same three constraint domains and tests group weighting, nonlinear aggregation, and success thresholds from one through ten pairs.
- We quantify the validity boundary of common remedies. We measure witness-ordering discordance, pair-rematching sensitivity, Clopper–Pearson versus Wilson sensitivity, threshold-dependent capability boundaries, and ideal versus residual shortcut-ceiling boundaries. The results show which failure modes controlled metrics correct and which conclusions remain conditional on benchmark construction, aggregation, and auxiliary evidence.

2 Limits of Prediction Metrics: Non-Identifiability and Non-Monotonicity

This section formalizes the central limitation of prediction-based evaluation. We focus on binary classification because many ML-based software-engineering tasks are ultimately evaluated as binary decisions: vulnerable or non-vulnerable, satisfiable or unsatisfiable, buggy or clean, and pass or fail. Our argument is not specific to any particular metric. Instead, we identify a class of metrics whose inputs are finite sequences of labels and predictions. We show two limitations of this class. First, such metrics cannot identify the mechanism behind a prediction when two models agree on benchmark outputs. Second, even when models differ, a higher metric score does not necessarily imply stronger reasoning.

2.1 Semantic Factors and Binary-Prediction Metrics

We conceptually decompose each benchmark instance x into a semantic factor z and non-semantic information c . The semantic factor contains the information that should determine the ground-truth label under the intended task semantics. Non-semantic information may be visible in the instance and may influence a model’s prediction, but it should not determine the label. This decomposition is conceptual and need not correspond to an observable partition of tokens, statements, or graph elements.

In our target domains, z captures the program relationships that determine whether a vulnerability is present or the logical relationships that determine whether a formula is satisfiable. The non-semantic component c captures presentation choices, construction artifacts, and other information that does not determine the label under the intended semantics. We use the term *side-channel cue* for non-semantic information that supports a model’s predictions on a particular benchmark.

Let a benchmark be

$$B = \{(x_i, y_i)\}_{i=1}^n.$$

For each instance, the ground-truth label is determined by its semantic factor:

$$y_i = g(z_i).$$

A model f observes the complete instance x_i and produces

$$\hat{y}_i = f(x_i), \quad i = 1, \dots, n.$$

The intended behavior is for $\hat{y}_i$ to depend on z_i . However, the model may also use information in c_i , alone or together with z_i .

We call M a *binary-prediction metric* if it depends only on the finite sequence of ground-truth labels and binary predictions:

$$M(B, f) = \Phi((y_1, \dots, y_n), (\hat{y}_1, \dots, \hat{y}_n)).$$

Here Φ denotes the metric-specific computation. For accuracy, Φ averages the indicators $\mathbf{1}[y_i = \hat{y}_i]$. For F1, it computes a function of true positives, false positives, and false negatives. For MCC, it computes the standard correlation coefficient from the confusion matrix. This definition also covers precision, recall, Cohen's Kappa, G-mean, Youden's J , and other metrics computed solely from the confusion matrix. It includes lower confidence bounds, such as Wilson lower confidence bounds, when they are applied to a binary success event derived from labels and predictions.

Some metrics additionally use benchmark-defined pair, group, or cluster structure. Let $\mathcal{S}$ describe this structure. A structured binary-prediction metric has the form

$$M(B, f, \mathcal{S}) = \Phi((y_1, \dots, y_n), (\hat{y}_1, \dots, \hat{y}_n), \mathcal{S}).$$

This class includes pair-level ADR, group-level ADR, cluster-level ADR, and related metrics that aggregate binary outcomes over controlled collections of instances. The structure $\mathcal{S}$ changes the behavioral criterion by adding relationships among benchmark instances, but the metric is still computed from labels, predictions, and that structure. Proposition 1 therefore applies to structured metrics as well. Section 4 examines when the stronger behavioral criterion provides better evidence of reasoning and why its scores still need not order models by reasoning ability.

2.2 Mechanism Non-Identifiability

The following proposition states the basic non-identifiability result.

Proposition 1: Mechanism non-identifiability. Let M be any metric that depends only on the finite vectors of ground-truth labels, model predictions, and optionally benchmark structure $\mathcal{S}$. Let f_r and f_s be two models such that

$$f_r(x_i) = f_s(x_i) \quad \text{for all } i = 1, \dots, n.$$

Then

$$M(B, f_r) = M(B, f_s).$$

Proof. Since f_r and f_s agree on all benchmark instances, they induce the same prediction vector:

$$(\hat{y}_1^r, \dots, \hat{y}_n^r) = (\hat{y}_1^s, \dots, \hat{y}_n^s).$$

The ground-truth labels and the benchmark structure $\mathcal{S}$ are the same for both models. Therefore, the inputs to M are identical, and M must assign the same value to both models. $\square$

This proposition is simple, but its implication is important. Suppose f_r follows the semantic factor z , while f_s relies on non-semantic information in c , such as a side-channel cue. If the two models agree on all benchmark predictions, every binary-prediction metric assigns them the same score. Equal metric values therefore do not establish equal reasoning ability, nor can the metric prefer the model that follows the intended semantic relationship.

This is the paper's basic non-identifiability observation. It is not specific to F1, MCC, Wilson bounds, or ADR; it follows from the finite prediction vectors on which these metrics are computed. The more consequential question for model comparison is whether a higher metric value implies stronger reasoning. The next subsection shows that it does not.

2.3 Higher Metric Scores Do Not Imply Stronger Reasoning

The previous proposition considers the case where two models agree on all benchmark predictions. One might argue that this case is too restrictive: in practice, two models often make different predictions, and the model with the higher metric score may appear to be better. We next show that even in this more general case, higher prediction performance does not necessarily imply stronger reasoning.

Proposition 2: Metric non-monotonicity with respect to reasoning. A shortcut model can receive a higher prediction-metric score than a more semantic model when the shortcut correlation yields fewer benchmark errors than the semantic model's imperfect use of semantic evidence.

Construction. Let $Y \in \{0, 1\}$ be the ground-truth label, and assume that the label is determined by a semantic factor Z . Consider a shortcut model f_c that predicts from a side-channel cue and a semantic model f_z that predicts from semantic evidence but does so imperfectly. For the shortcut model, define the class-conditional error rates

$$\epsilon_1 = \Pr[f_c(X) = 0 \mid Y = 1], \quad \epsilon_2 = \Pr[f_c(X) = 1 \mid Y = 0].$$

Thus,

$$\text{TPR}_c = 1 - \epsilon_1, \quad \text{TNR}_c = 1 - \epsilon_2.$$

For the semantic model, define

$$\delta_1 = \Pr[f_z(X) = 0 \mid Y = 1], \quad \delta_2 = \Pr[f_z(X) = 1 \mid Y = 0],$$

so that

$$\text{TPR}_z = 1 - \delta_1, \quad \text{TNR}_z = 1 - \delta_2.$$

Let

$$\pi = \Pr[Y = 1]$$

be the positive-class prevalence. The shortcut model has accuracy

$$\text{Acc}_c = 1 - (\pi\epsilon_1 + (1 - \pi)\epsilon_2),$$

whereas the semantic model has accuracy

$$\text{Acc}_z = 1 - (\pi\delta_1 + (1 - \pi)\delta_2).$$

Therefore,

$$\text{Acc}_c > \text{Acc}_z$$

if and only if

$$\pi\epsilon_1 + (1 - \pi)\epsilon_2 < \pi\delta_1 + (1 - \pi)\delta_2.$$

In words, the shortcut model receives higher accuracy whenever its prevalence-weighted benchmark error is smaller than that of the semantic model. In the special case where errors are symmetric across the two classes, so that

$$\epsilon_1 = \epsilon_2 = \epsilon \quad \text{and} \quad \delta_1 = \delta_2 = \delta,$$

this condition reduces to

$$\epsilon < \delta.$$

Thus, in the symmetric case, the shortcut model receives higher accuracy exactly when its class-independent error rate is smaller than that of the semantic model.

The same comparison can be written for other confusion-matrix metrics. Balanced accuracy gives

$$\text{BAcc}_c = 1 - \frac{\epsilon_1 + \epsilon_2}{2}, \quad \text{BAcc}_z = 1 - \frac{\delta_1 + \delta_2}{2}.$$

Thus,

$$\text{BAcc}_c > \text{BAcc}_z$$

if and only if

$$\epsilon_1 + \epsilon_2 < \delta_1 + \delta_2.$$

The same condition applies to Youden's J , since

$$J_c = \text{TPR}_c + \text{TNR}_c - 1 = 1 - \epsilon_1 - \epsilon_2.$$

For G-mean,

$$\text{G-mean}_c = \sqrt{(1 - \epsilon_1)(1 - \epsilon_2)}, \quad \text{G-mean}_z = \sqrt{(1 - \delta_1)(1 - \delta_2)}.$$

Hence the shortcut model receives a higher G-mean exactly when

$$(1 - \epsilon_1)(1 - \epsilon_2) > (1 - \delta_1)(1 - \delta_2).$$

For F1, the shortcut model obtains

$$\text{F1}_c = \frac{2\pi(1 - \epsilon_1)}{2\pi(1 - \epsilon_1) + (1 - \pi)\epsilon_2 + \pi\epsilon_1},$$

and the semantic model obtains the analogous expression with δ_1, δ_2 replacing ϵ_1, ϵ_2 . Thus F1 can also rank the shortcut model higher whenever the shortcut errors produce a more favorable positive-class confusion profile on the benchmark.

This proposition does not show that the shortcut model is better at reasoning. It shows that prediction metrics rank models according to benchmark label agreement. If a shortcut cue yields fewer benchmark errors than the model's imperfect use of semantic evidence, the metric will rank the shortcut model higher.

The same error-rate view also explains why reporting multiple metrics does not solve the problem. Multiple metrics provide more information about a model's error profile, but they do not make metric orderings monotone with reasoning ability. If

$$\epsilon_1 < \delta_1 \quad \text{and} \quad \epsilon_2 < \delta_2,$$

then the shortcut model makes fewer errors on both positive and negative instances. Any metric that is monotone in true-positive rate and true-negative rate will then rank the shortcut model above the semantic model. This includes accuracy, balanced accuracy, G-mean, Youden's J , and many other confusion-matrix metrics. Thus, even unanimous agreement among multiple prediction metrics does not imply stronger reasoning ability. It may only show that the shortcut model has a more favorable benchmark error profile.

When the error profiles cross, multiple metrics can reveal useful tradeoffs. For example, the shortcut model may have a lower false-negative rate but a higher false-positive rate. In that case, different metrics may disagree because they weight class prevalence, positive-class performance, and negative-class performance differently. Reporting multiple metrics can therefore shrink the set of shortcut error profiles that appear better than the semantic model. However, it cannot make the joint metric ordering a reliable ordering of reasoning ability. If a shortcut model satisfies the relevant inequalities for all reported metrics, the metrics will jointly favor it even though it does not capture the intended semantic relationship. The value of multiple metrics is diagnostic of benchmark behavior, not a direct measure of reasoning.

The analysis so far concerns metrics computed from binary predictions. One possible response is to retain the scalar scores from which those predictions are derived. Score-based metrics avoid dependence on a particular classification threshold and preserve ranking information that binary predictions discard. Section 3 examines whether this additional information makes metric values more closely aligned with reasoning ability.

A different response is to change the evaluation design by organizing instances into controlled pairs, groups, or clusters. As noted in Section 2.1, these metrics remain subject to Proposition 1, but their stronger behavioral criteria can make some non-semantic strategies less effective. Section 4 analyzes their benefits, calibration tradeoffs, and remaining limitations.

The results are independent of model architecture. Section S6 of the supplementary material provides illustrative instantiations for additive classifiers, attention-based models, and GNNs.

3 Why Score-Based Metrics Such as AUROC Still Fall Short

The preceding section considered metrics computed from binary predictions. One might hope that score-based metrics provide a better measure of reasoning ability because they retain more information than thresholded predictions. AUROC is a representative example. Rather than evaluating the binary predictions obtained at a fixed threshold, AUROC evaluates whether positive instances receive higher scores than negative instances. It therefore avoids the information loss introduced by

$$s_f(x) \mapsto 1[s_f(x) \geq \tau].$$

This additional information improves the measurement of ranking quality, but it does not establish a monotone relationship between metric values and reasoning ability. A model can obtain a high AUROC by exploiting a non-semantic cue that separates the benchmark classes effectively. It can also obtain a higher AUROC than a model that attempts to capture the intended semantic relationship but does so imperfectly. Thus, a higher AUROC indicates better ranking performance on the evaluated distribution, not necessarily stronger reasoning.

3.1 Score-Based Metrics

Let a model f produce a scalar score

$$s_i = s_f(x_i),$$

where larger scores indicate a stronger preference for the positive class. A score-based metric has the form

$$M_{\text{score}}(B, f) = \Psi((y_1, \dots, y_n), (s_1, \dots, s_n)).$$

The function Ψ may depend on the numerical score values or only on the ranking they induce.

AUROC is a rank-based metric in this class. At the population level, it can be written as

$$\text{AUROC} = \Pr[s_f(X^+) > s_f(X^-)] + \frac{1}{2} \Pr[s_f(X^+) = s_f(X^-)],$$

where X^+ is drawn from the positive class and X^- is drawn from the negative class. AUROC therefore measures how frequently the scoring function ranks a positive instance above a negative instance, with half credit for ties.

This definition gives AUROC an appropriate and useful interpretation as a measure of benchmark ranking quality. It does not, however, justify interpreting AUROC as a measure of reasoning ability. Two scoring functions that induce the same benchmark ordering receive the same AUROC even when one relies on the intended semantic relationship and the other relies on non-semantic cues. Thus, AUROC remains subject to the non-identifiability limitation established for prediction metrics. More importantly, as the next construction shows, AUROC also remains non-monotone with respect to reasoning ability.

3.2 A Shortcut Construction for AUROC

We instantiate the limitation using the same semantic and non-semantic setup as in the previous sections. Let $Y \in \{0, 1\}$ be the ground-truth label, which is determined by the semantic factor. Let

$C \in \{0, 1\}$ be a non-semantic cue satisfying

$$\Pr[C = 1 \mid Y = 1] = a$$

and

$$\Pr[C = 1 \mid Y = 0] = b,$$

where $a > b$. Consider the shortcut scoring function

$$s_c(x) = C(x).$$

This scoring function ranks instances using only the non-semantic cue.

For a randomly drawn positive instance X^+ and negative instance X^- , the resulting AUROC is

$$\text{AUROC}_c = \Pr[C(X^+) > C(X^-)] + \frac{1}{2} \Pr[C(X^+) = C(X^-)].$$

Because C is binary,

$$\Pr[C(X^+) > C(X^-)] = a(1 - b),$$

and

$$\Pr[C(X^+) = C(X^-)] = ab + (1 - a)(1 - b).$$

Therefore,

$$\begin{aligned} \text{AUROC}_c &= a(1 - b) + \frac{1}{2}(ab + (1 - a)(1 - b)) \\ &= \frac{1}{2} + \frac{1}{2}(a - b). \end{aligned}$$

For example, if $a = 0.86$ and $b = 0.10$, then $\text{AUROC}_c = 0.88$. The shortcut scoring function therefore achieves excellent ranking performance even though it does not capture the semantic relationship that determines the labels.

This construction illustrates why a high AUROC should not be interpreted as evidence of strong reasoning. AUROC rewards separation between positive and negative benchmark instances. A non-semantic cue can produce that separation when it is strongly correlated with the labels. The resulting AUROC correctly describes the ranking induced on the benchmark, but it overstates what can be concluded about the model's reasoning ability.

The construction also gives a score-based instance of metric non-monotonicity. Let s_z be a scoring function that attempts to capture the semantic relationship between instances and labels, but does so imperfectly. If

$$\frac{1}{2} + \frac{1}{2}(a - b) > \text{AUROC}_z,$$

then AUROC ranks the shortcut scoring function above the semantic scoring function. The shortcut function receives the higher metric value because its non-semantic cue produces a more favorable ranking on the benchmark. The higher value does not imply that the shortcut function has greater reasoning ability.

The same conclusion is not specific to the binary cue used in this construction. Any non-semantic score that orders positive and negative benchmark instances more reliably than an imperfect semantic score can receive the higher AUROC. The metric orders models according to their observed ranking performance, not necessarily according to how successfully they capture the intended semantic relationship.

3.3 Implications for Reasoning-Oriented Evaluation

The limitation of AUROC does not make it inappropriate for measuring ranking quality. AUROC is useful when a system is intended to prioritize instances for human inspection, such as ranking suspicious functions for vulnerability review. A higher AUROC indicates that the scoring function better orders positive and negative instances on the evaluated distribution. This is a meaningful behavioral comparison. It should not automatically be extended into a reasoning comparison between the models.

Reasoning-oriented evaluation therefore requires evidence beyond aggregate scores and rankings. Controlled interventions can test whether scores respond to the intended semantic relationships. For example, an evaluation can change the semantic factor z while holding non-semantic information c as fixed as possible. Matched SAT/UNSAT formulas and vulnerable/fixed program variants follow this design. Complementary interventions can vary c while preserving z to test whether the ranking is unnecessarily sensitive to non-semantic changes.

Such interventions make score-based evaluation more informative, but they do not convert AUROC or a controlled ranking metric into a direct measure of reasoning ability. As shown in Section 4, controlled outcomes can still combine semantic success with residual non-semantic success, and the two contributions remain non-separable. A higher controlled score therefore provides stronger performance under the chosen evaluation design, not conclusive evidence of stronger reasoning.

Score-based metrics should consequently be interpreted together with task-specific evidence. When mechanically checkable witnesses are available, such as satisfying assignments in SAT, they provide stronger validation that the model has produced a semantically valid result. Vulnerability detection can also admit witnesses, including vulnerable slices, execution traces, or triggering inputs. Their validation is more context dependent and generally harder to automate. When strong witness validation is unavailable or difficult, metamorphic relations and behavioral perturbation tests [3, 26] provide weaker but still useful evidence.

In summary, AUROC retains more behavioral information than a thresholded classification metric and is valuable for evaluating ranking quality. It nevertheless suffers from both non-identifiability and non-monotonicity with respect to reasoning ability. A higher AUROC establishes a better benchmark ranking. It does not, without additional evidence, establish stronger semantic reasoning.

4 What Controlled Metrics Fix and What They Still Miss

The preceding sections considered two ways of summarizing model behavior: metrics computed from binary predictions and metrics computed from scalar scores or rankings. Retaining scores avoids threshold dependence, but neither form of evaluation guarantees that a higher metric value corresponds to stronger reasoning. A different response is to change the evaluation design rather than only the metric formula.

Pair, group, and cluster metrics evaluate controlled collections of related instances rather than independently sampled predictions. Their goal is to reduce the usefulness of non-semantic cues and make measured success more closely aligned with semantic success. These designs can provide stronger reasoning-oriented evidence than ordinary benchmark evaluation. This section analyzes what they improve, the calibration tradeoffs they introduce, and why their observed success rates still do not uniquely determine semantic success.

4.1 Mixed Mechanisms and the Target Quantity

The pure semantic and pure shortcut models used in the previous section are idealized endpoints. Practical ML systems may obtain correct outcomes through semantic information, a residual side-channel cue, lucky guessing, or a combination of these sources. For evaluation, the central question is how the observed success rate relates to semantic success.

We use a simple mixed-success abstraction to describe this case. For an evaluation unit U , let $E_{\text{sem}}(U)$ be the event of semantic success. Let $E_{\text{short}}(U)$ be the event that a residual non-semantic cue or lucky guess produces success when semantic success does not occur. Define

$$\delta_U = \Pr[E_{\text{sem}}(U)], \quad \epsilon_U = \Pr[E_{\text{short}}(U) \mid \neg E_{\text{sem}}(U)].$$

Then the observed success probability for U is

$$p_U = \delta_U + (1 - \delta_U)\epsilon_U. \quad (1)$$

Thus, the gap between observed success and semantic-only success is

$$p_U - \delta_U = (1 - \delta_U)\epsilon_U.$$

Equation (1) captures the goal and limitation of controlled metrics. Ideally, the metric would measure δ_U , the probability of semantic success. In reality, it observes a success event that may also include residual non-semantic contribution. Controlled metrics therefore try to make ϵ_U small by constructing evaluation units in which simple non-semantic strategies are less useful. They succeed only to the extent that residual shortcut or guessing success is actually reduced.

4.2 Pair Metrics

A matched pair is an evaluation unit

$$U_{\text{pair}} = (x^+, x^-)$$

containing two instances whose labels differ because the semantic factor changes. Ideally, pair construction would change only the semantic factor while keeping non-semantic information fixed. In software-engineering tasks, however, a semantic edit often introduces lexical or syntactic differences that can themselves become predictive cues. For example, in vulnerability detection, a non-vulnerable variant of a vulnerable program may be created by adding a condition guard. Although the edit may be small relative to the whole function, it can introduce tokens such as `if`, relational operators, parentheses, braces, and semicolons. These tokens may become predictive artifacts of the repair. Consequently, correct pair classification is not by itself equivalent to semantic success.

We formalize this residual difference as follows. Let z_1 and c^+ denote the semantic and non-semantic components of x^+ , and let z_0 and c^- denote those of x^- . The semantic components satisfy

$$g(z_1) = 1, \quad g(z_0) = 0,$$

while the non-semantic components may differ:

$$c^- = c^+ \oplus \Delta_c(z_1, z_0).$$

Here $\Delta_c(z_1, z_0)$ denotes the non-semantic differences induced by the semantic change from z_1 to z_0 . The goal of pair construction is to make this residual difference small, not to assume that it is zero.

A pair-level metric such as ADR counts the pair as successful when the model predicts both instances correctly:

$$f(x^+) = 1 \quad \text{and} \quad f(x^-) = 0.$$

Instantiating Equation (1) for a pair gives

$$p_{\text{pair}} = \delta_{\text{pair}} + (1 - \delta_{\text{pair}})\epsilon_{\text{pair}}, \quad (2)$$

where δ_{pair} is the semantic pair-success probability, and ϵ_{pair} is the residual pair-success probability from side-channel cues or lucky guessing when semantic success does not occur. Pair ADR estimates p_{pair} , not δ_{pair} directly.

The benefit of pair construction is that it can reduce ϵ_{pair} . For example, a deterministic model that predicts only from side-channel information shared by the two instances will tend to give the same prediction to both and therefore fail the pair. However, if $\Delta_c(z_1, z_0)$ contains predictive artifacts induced by the semantic edit, those artifacts can increase pair success without a corresponding increase in semantic ability. Pair metrics therefore move the measured success rate closer to semantic success only to the extent that the residual side-channel component is small. A higher pair ADR is consequently more informative than marginal accuracy alone, but it is not automatically evidence of stronger reasoning.

4.3 Group Metrics

Group metrics ask whether pair-level success persists across a controlled neighborhood. Let $\mathcal{T}(x)$ denote transformations that preserve the task-relevant structure within one arm of a matched construction. A model that tracks that structure should be stable over $\mathcal{T}(x)$, just as object identity should survive rotations or crops that preserve the defining object. For constraint problems, each group contains several positive–negative pairs formed around a common generator instance. Within-arm variations retain the decision-relevant structure, whereas the controlled positive–negative edit changes the correct decision. Repeated pair success tests both requirements: invariance to incidental variation and sensitivity to the decisive edit.

This interpretation makes group success stronger behavioral evidence than an isolated correct pair, but it does not identify the mechanism that produced the labels. Correlated group-level cues can also remain stable, and a label-only score does not validate a constructive witness. Group consistency and witness validity are therefore complementary evidence rather than substitutes.

We first define the continuous group aggregations used in our experiments. Let $\mathcal{G}_c$ be the groups containing at least one complete matched pair. For group g , let n_g be the number of complete pairs, S_g the number for which both labels are correct, and $a_g = S_g/n_g$ the within-group ADR. Equal-group and equal-pair aggregation give

$$\text{GADR} = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} a_g, \quad \text{ADR} = \frac{\sum_{g \in \mathcal{G}_c} S_g}{\sum_{g \in \mathcal{G}_c} n_g}. \quad (3)$$

GADR gives every evaluable group equal weight; ADR gives every complete pair equal weight. Their difference is a weighting effect, not evidence that one aggregation recovers the latent reasoning mechanism.

To expose within-group uncertainty, define the Bernoulli variance $v_g = a_g(1 - a_g)$. For $\lambda \geq 0$, we use

$$\text{SGADR}_\lambda = \frac{1}{|\mathcal{G}_c|} \sum_{g \in \mathcal{G}_c} [a_g - \lambda v_g], \quad \text{SGADR}_{\star, \lambda} = \text{ADR} - \lambda \text{ADR}(1 - \text{ADR}). \quad (4)$$

The first expression applies the nonlinear transformation before macro aggregation; the second applies it after micro aggregation. At $\lambda = 1$,

$$\text{SGADR}_1 = \frac{1}{|\mathcal{G}_c|} \sum_g a_g^2, \quad \text{SGADR}_{\star, 1} = \text{ADR}^2. \quad (5)$$

We also report the between-group variance $\frac{1}{|\mathcal{G}_c|} \sum_g (a_g - \text{GADR})^2$. When all groups have equal complete-pair counts, ADR equals GADR and the difference $\text{SGADR}_1 - \text{SGADR}_{\star,1}$ equals this variance. Large separation therefore localizes success in a subset of groups, but does not explain why those groups were successful.

Group metrics can further strengthen pair metrics by applying a repeated-success criterion. We reuse the pair-level quantities from Section 4.2. Pair ADR estimates p_{pair} in Equation (2), while the quantity we would like to approximate for reasoning ability is δ_{pair} . Since $p_{\text{pair}} \geq \delta_{\text{pair}}$, a group threshold can be viewed as an attempt to calibrate the observed pair success rate downward by requiring repeated success.

Suppose there are m groups, each containing n matched positive–negative pairs. Let S_j be the number of successful pairs in group j . The group is counted as successful when

$$S_j \geq k.$$

The corresponding group pass rate is

$$A_{\geq k} = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[S_j \geq k]. \quad (6)$$

Under a simplifying reference model in which pair successes are independent and have common success probability p , the probability that a group passes the threshold is

$$T_{n,k}(p) = \sum_{r=k}^n \binom{n}{r} p^r (1-p)^{n-r}.$$

Applying this reference model with $p = p_{\text{pair}}$, the expected group score is

$$\mathbb{E}[A_{\geq k}] = T_{n,k}(p_{\text{pair}}).$$

Thus, $A_{\geq k}$ estimates a threshold-transformed mixed success probability, not p_{pair} directly.

The desired calibration property of group ADR is therefore

$$T_{n,k}(p_{\text{pair}}) \approx \delta_{\text{pair}}.$$

The approximation error relative to the semantic target is

$$T_{n,k}(p_{\text{pair}}) - \delta_{\text{pair}}.$$

For analysis, this error can be decomposed as

$$T_{n,k}(p_{\text{pair}}) - \delta_{\text{pair}} = \underbrace{T_{n,k}(p_{\text{pair}}) - T_{n,k}(\delta_{\text{pair}})}_{\text{residual non-semantic contribution}} + \underbrace{T_{n,k}(\delta_{\text{pair}}) - \delta_{\text{pair}}}_{\text{threshold-induced bias}}.$$

This decomposition is conceptual rather than directly observable: δ_{pair} and ϵ_{pair} are latent, so the two terms cannot in general be estimated from $A_{\geq k}$ alone. Nevertheless, it clarifies the calibration tradeoff. The first term is nonnegative and captures the remaining inflation caused by residual shortcuts or lucky successes. The second term is introduced by the group threshold and may be positive or negative.

Choosing k is therefore difficult. If k is too small, the group score may remain inflated by residual non-semantic success:

$$T_{n,k}(p_{\text{pair}}) > \delta_{\text{pair}}.$$

If k is too large, the group criterion may over-penalize imperfect semantic behavior:

$$T_{n,k}(p_{\text{pair}}) < \delta_{\text{pair}}. \quad (7)$$

This over-penalization can occur even though $p_{\text{pair}} \geq \delta_{\text{pair}}$, because group ADR does not observe the latent semantic target δ_{pair} . It observes only whether each group passes the threshold.

For fixed n , p_{pair} , and δ_{pair} , the over-penalizing thresholds are those satisfying Equation (7). This set can be written as

$$K_{\text{over}} = \{k \in \{1, \dots, n\} : T_{n,k}(p_{\text{pair}}) < \delta_{\text{pair}}\}.$$

Since $T_{n,k}(p_{\text{pair}})$ is nonincreasing in k , once a threshold satisfies Equation (7), every stricter threshold also satisfies it. Intuitively, when the required success fraction k/n is substantially above p_{pair} , the threshold requires more successful pairs than the model obtains on average, and the group score can fall below δ_{pair} .

The all-pairs threshold is an extreme example. When $k = n$,

$$T_{n,n}(p_{\text{pair}}) = p_{\text{pair}}^n.$$

Therefore, the all-pairs group score underestimates the pair-level semantic target exactly when

$$p_{\text{pair}}^n < \delta_{\text{pair}}.$$

From Equation (2), $p_{\text{pair}} < 1$ unless semantic success is perfect ($\delta_{\text{pair}} = 1$) or residual non-semantic success is perfect whenever semantic success fails ($\epsilon_{\text{pair}} = 1$). Therefore, when $0 < \delta_{\text{pair}} < 1$ and $p_{\text{pair}} < 1$, p_{pair}^n decreases toward 0 as n increases. For a sufficiently large n , the all-pairs threshold will understate the pair-level semantic target unless the mixed pair-success probability is perfect.

The shortcut-only case shows the opposite benefit. If $\delta_{\text{pair}} = 0$, then group success due only to residual side-channel cues or lucky guessing is $T_{n,k}(\epsilon_{\text{pair}})$. For small ϵ_{pair} and $k \geq 2$, this quantity has leading order ϵ_{pair}^k , up to the binomial coefficient. Thus, group ADR can substantially attenuate weak independent shortcut or guessing effects.

Overall, group ADR is an indirect calibration attempt. Increasing k can suppress weak independent shortcut or guessing effects, but a threshold that is too strict can also drive the measured group score below the latent pair-level semantic target. The choice of k therefore depends on the residual shortcut rate, the difficulty distribution, the construction of the groups, and the dependence structure among pairs. Group ADR can move measured success closer to semantic success in a particular controlled setting, but it does not directly measure semantic success.

4.4 Cluster Metrics

Group metrics reduce weak shortcut effects by requiring repeated pair-level success, but they can introduce a matching instability: the score may depend on how instances are paired within a group. Suppose a group contains positive instances $P_1, \dots, P_n$ and negative instances $N_1, \dots, N_n$ that are intended to test the same underlying capability. A group ADR score first forms matched positive-negative pairs and then counts how many pairs are successful. For a matching π , the number of successful pairs is

$$S_\pi = \sum_{i=1}^n \mathbf{1}[f(P_i) = 1] \mathbf{1}[f(N_{\pi(i)}) = 0].$$

The value of S_π may change when the matching π changes. Thus, two evaluators can obtain different group ADR values from the same set of model predictions merely by pairing instances differently. This is undesirable when the group is intended to measure a shared capability rather than a particular arbitrary matching.

Cluster metrics aim to avoid this matching instability by evaluating the collection as a whole rather than as a set of repaired pairs. For example, a cluster may contain one positive anchor X^+

and multiple negative variants $X_1^-, \dots, X_n^-$. The cluster is counted as successful if the anchor is classified correctly and at least k negative variants are classified correctly:

$$f(X^+) = 1 \quad \text{and} \quad \sum_{i=1}^n \mathbf{1}[f(X_i^-) = 0] \geq k.$$

This criterion depends only on the predictions within the cluster, not on an arbitrary matching between positives and negatives. In this sense, cluster metrics replace edge-level pair success with set-level or cluster-level success.

However, cluster metrics still remain prediction-output metrics. They can make the score less sensitive to arbitrary pair construction, and they can reduce weak shortcut effects by requiring repeated success across variants. A higher cluster score therefore indicates greater consistency under the cluster criterion, but correlated residual cues can improve that score without a corresponding increase in semantic ability. In the mixed-success view, a cluster score is still a threshold-transformed mixed success probability, not a direct estimate of the semantic component.

4.5 Limitations of Controlled Prediction Metrics

The preceding analysis shows that controlled metrics are stronger evaluation designs, but not certificates of reasoning. Pair metrics reduce the usefulness of non-semantic cues that are shared across opposite-label instances, but semantic edits can still introduce residual artifacts. Group metrics can attenuate weak independent shortcut success by requiring repeated pair-level success, but choosing k is a calibration problem. A threshold that is too small may leave shortcut success influential, while one that is too large may understate imperfect semantic ability. Cluster metrics address matching instability, but they still observe only cluster-level prediction success.

The calibration problem also limits the possibility of turning group or cluster ADR into a portable metric framework by experimentally tuning its aggregation rule. Even if additional evidence were available to tune the rule on one dataset, the resulting calibration would depend on that dataset's residual shortcut rate, difficulty distribution, controlled-instance construction, and dependence structure. These properties can change across groups, benchmarks, model families, and problem domains. A rule that makes the measured score close to semantic success in one setting may leave shortcut effects influential in another, or may over-penalize imperfect semantic behavior on a harder benchmark.

These observations lead to a third general limitation.

Proposition 3: Non-separability of semantic and non-semantic success. Let U be an evaluation unit whose observed success probability satisfies Equation (1). For any $0 < p_U < 1$, the observed value p_U does not uniquely determine the semantic success probability δ_U and the residual non-semantic success probability ϵ_U . Infinitely many pairs (δ_U, ϵ_U) are compatible with the same observed success probability.

Proof. Equation (1) provides only one constraint on the two latent quantities δ_U and ϵ_U . For every choice of $\delta_U \in [0, p_U]$, solving the equation for ϵ_U yields a valid value in $[0, 1]$, so infinitely many pairs (δ_U, ϵ_U) produce the same observed value p_U . $\square$

Proposition 3 complements the two limitations established in Section 2. Proposition 1 shows that equal metric values can accompany different reasoning behavior, and Proposition 2 shows that a higher metric value need not imply stronger reasoning. Non-separability adds that even exact knowledge of a controlled unit's success probability does not determine its semantic success probability. A finite estimate or deterministic transformation of p_U cannot provide the missing ordering of reasoning ability.

This limitation is especially important when residual shortcut successes are correlated within a pair, group, or cluster. For example, a group-level artifact that causes all pairs in a group to be classified correctly with probability ϵ_{pair} can pass a group threshold with the same probability, regardless of k . Controlled metrics can make some shortcuts less useful and can move measured success closer to semantic success in a particular controlled setting. A higher controlled score is therefore stronger behavioral evidence, but it does not by itself establish stronger reasoning. Additional evidence, such as witnesses, semantic interventions, and non-semantic perturbation tests, is still needed.

5 Experimental Design

The formal results establish what finite prediction records cannot identify in general. The experiments have a narrower purpose: they test whether the corresponding failures occur in realistic matched evaluations, what stronger observable criteria reveal, and which ambiguities remain after the evaluation unit is strengthened. The strongest archived reference is witness-aware ADR: it requires the correct matched labels and a positive-side artifact accepted by an independent checker. We therefore use W as a *witness-grounded proxy for demonstrated reasoning ability*. It is closer to the intended construct than a label-only metric because credited success includes a task-semantic construction, and it is conservative because the protocol does not observe every reasoning process that a model may perform. Neither W , a high matched score, nor a positive shortcut-adjusted margin is treated as an unconditional measure of the model's complete latent reasoning ability.

5.1 Research Questions

The empirical study is organized around three questions.

RQ1. How far are conventional, matched, and confidence-bounded metrics from witness-grounded reasoning evidence, in both level and model ordering?

RQ2. What failures are exposed when label agreement is separated from two-sided label recognition, positive-side witness production, and available two-sided witness evidence?

RQ3. How sensitive are controlled conclusions to pair matching, group weighting, nonlinear aggregation, repeated-success thresholds, confidence construction, the reporting threshold, and the specified shortcut adversary and ceiling?

RQ1 operationalizes the non-monotonicity result of Proposition 2 by comparing metric-induced model orderings with a stronger observable reference. RQ2 asks what information is gained by strengthening the evidence unit from an individual label to a matched contrast and then to validated artifacts. RQ3 examines the construction dependence characterized by Proposition 3.

5.2 Frozen Archives and Source Separation

We reanalyze two primary frozen experimental archives from prior matched and shortcut-aware studies and perform a separate group analysis from the current raw prediction records. The *matched-separability archive* contains complete per-size tables for 3-SAT, 2-SAT, and graph 3-coloring. It reports ordinary category metrics, ADR, positive-side witness checks, a group-averaged two-sided point estimate, Clopper–Pearson and Wilson lower bounds, and fixed, maximum, and minimum pair-rematching variants. Across the three constraint domains it contains 19 3-SAT, 59 2-SAT, and 45 graph-coloring model–size rows, for 123 rows in total. The refreshed 2-SAT export additionally contains a follow-up audit requiring both a verified SAT assignment and a verified UNSAT clause subset.

The *shortcut-margin archive* contains 3-SAT, 2-SAT, and graph coloring under an explicit structural shortcut account. Its nested design uses 11 matched positive/negative pairs per group. For each

Table 1. Frozen experimental archives and their use in this paper. Overlapping rows are never averaged or joined across archives because refreshed provider runs, parsing rules, and follow-up availability can produce version-specific values.

Archive	Domains and observables	Role and claim limit
Matched separability	3-SAT, 2-SAT, graph coloring; labels, ADR, positive witnesses, CP/Wilson bounds, pair-rematching branches, and a 2-SAT two-sided witness follow-up	Tests metric-witness level and ordering disagreement, one-sided behavior, label-without-witness behavior, confidence-method sensitivity, matching sensitivity, and threshold-indexed boundaries. The quantities remain output-level criteria.
Shortcut-adjusted margin	3-SAT, 2-SAT, graph coloring; factorized lower bounds, an ideal 0.25 floor, and a permutation-calibrated structural shortcut ceiling	Tests how conclusions change when a specified nuisance-factor account is charged and when a lower confidence bound replaces a point estimate. A positive margin is relative only to the stated adversary battery; the raw-versus-residualized graph-coloring contrast is excluded because its frozen runner configurations are not distinct.
Raw-prediction group analysis	3-SAT, 2-SAT, and graph coloring; 130 model-size slices and 15 overall model rows, reporting GADR, ADR, SGADR $_{\lambda}$, SGADR $_{\star, \lambda}$, group pass rates for $k = 1, \dots, 10$, variance terms, completion counts, and witness-aware counterparts computed with each domain's validator	Tests equal-group versus equal-pair weighting, nonlinear aggregation before versus after pooling, repeated-success sensitivity, stability across controlled variants, and how much group-level consistency survives a witness requirement. It is analyzed separately from the matched-separability archive.

3-SAT size, it uses 10 groups, yielding 220 labeled instances per size and 1,320 nominal instances over six sizes. For each 2-SAT or graph-coloring size, it uses 20 groups, yielding 440 instances per size; the nominal totals are 4,400 over ten 2-SAT sizes and 2,640 over six graph-coloring sizes, before provider, parsing, or artifact-availability exclusions. We use the graph-coloring rows only for analyses that do not compare the raw and complexity-residualized shortcut branches, because the frozen runner records the same configuration for both labeled exports.

The *raw-prediction group analysis* uses the established parsing and deduplication rules for each domain, then applies one domain-neutral aggregation implementation to the completed pair outcomes. It contains 19 3-SAT, 66 2-SAT, and 45 graph-coloring model-size slices, for 130 slices in total, plus one overall row for each of the 15 model configurations. Each group contains at most 11 candidate pairs. We report $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$, use $\lambda = 1$ for the main table, and compute $A_{\geq k}$ for every $k \in \{1, \dots, 10\}$. Every quantity is also computed in a witness-aware form that credits a pair only when the positive-side response carries a verified satisfying assignment (SAT) or proper coloring (graph coloring), using the same validators as the matched-separability archive.

The three constraint data sources overlap in tasks and model families, but several rows differ because they originate from different frozen reruns, completion patterns, parsing revisions, or reporting pipelines. We do not silently reconcile these values. All row-level metric, witness, rematching, and ordering analyses before Section 6.3 use the matched-separability archive. The shortcut-margin archive is used only for its internally consistent point-versus-bound and ideal-versus-residual ceiling analyses. The raw-prediction group outputs are used only for the aggregation and repeated-success analysis in Section 6.3. The 3-SAT and graph-coloring recomputations reproduce their archived group summaries. For 2-SAT, the current predictions and the older summary differ in 57 exported cells across four models; recomputation with both the new domain-neutral aggregator and the repository’s original 2-SAT aggregator agrees exactly on the current inputs. We therefore report the current-input values and retain the archived differences as a provenance audit rather than silently substituting either version. The witness-aware group columns reproduce the archived 3-SAT and graph-coloring witness summaries exactly, but not the archived 2-SAT witness values for Llama-Nemotron-Super-49B, QwQ-32B, Qwen3-14B, and Qwen3-32B. For these four models the current prediction files, and the frozen snapshot shipped with the shortcut-margin archive, contain many responses whose reasoning was cut off before the final answer object; the label parser recovers SAT/UNSAT from the free text, but there is no assignment to verify. Between 24% and 90% of their positives predicted positive therefore carry no witness, and the archived 2-SAT witness values cannot be reproduced from any prediction file that still exists. We report the current-input witness values, mark the four rows, and read them as lower bounds. This source separation prevents a conclusion assembled from incompatible experimental states.

5.3 Tasks, Models, and Evaluation Units

The constraint experiments use three matched decision problems. In 3-SAT and 2-SAT, a positive formula is satisfiable and its minimally edited counterpart is unsatisfiable. In graph 3-coloring, a positive graph is colorable and its minimally edited counterpart is non-colorable. The commercial 3-SAT summaries cover GPT-5, Claude Opus 4.7 at medium reasoning effort, and DeepSeek-Reasoner. The open-weight 2-SAT and graph-coloring summaries cover Qwen3-14B and Qwen3-32B in thinking mode, QwQ-32B, gpt-oss-20b in reasoning mode, Nemotron-H-47B-Reasoning, and Llama-Nemotron-Super-49B in reasoning mode. The full constraint archive contains 24 evaluated model configurations when provider and reasoning-mode variants are counted separately.

Boundaries are compared only within the same task family. In particular, 2-SAT is polynomial-time decidable, whereas 3-SAT is NP-complete, and the formula and graph generators induce different observable difficulty scales. A larger boundary on 2-SAT therefore does not establish greater reasoning ability than a smaller boundary on 3-SAT or graph coloring.

5.4 Observable Criteria

Instance- and pair-level category outcomes. For every model–size slice, we retain accuracy, per-class F1, G-mean, balanced accuracy, MCC, and the two arm recalls. For a matched group G , let

$$\hat{\theta}_+(G) = \frac{1}{|P_G|} \sum_{x \in P_G} \mathbf{1}[f(x) = 1], \quad \hat{\theta}_-(G) = \frac{1}{|N_G|} \sum_{x \in N_G} \mathbf{1}[f(x) = 0].$$

The group-level two-sided product is

$$s_G = \hat{\theta}_+(G) \hat{\theta}_-(G),$$

and the matched-separability point estimate at size N is the macro-average

$$\widehat{s}(N) = \frac{1}{|\mathcal{G}_N|} \sum_{G \in \mathcal{G}_N} s_G.$$

For comparison, the pooled product $s_{\text{direct}} = R_+ R_-$ equals the square of pooled G-mean. The group-averaged $\widehat{s}$ need not equal this pooled quantity because it preserves the internal matched-group partition. We use $\underline{s}^{\text{CP}}$ for the source archive's factorized Clopper–Pearson lower bound and $\underline{s}^W$ for its Wilson robustness check.

For an originally matched pair (x_i^+, x_i^-) , ADR records correct joint category recognition:

$$\text{ADR} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[f(x_i^+) = 1 \wedge f(x_i^-) = 0].$$

ADR is stricter than marginal accuracy but remains a label-only observable.

Witness-aware evidence. The positive-side witness reference is

$$W = \begin{cases} \text{ADR}^{+w}, & \text{for SAT,} \\ \text{ADR}^{+a}, & \text{for graph coloring.} \end{cases}$$

A SAT-side success receives witness credit only when the emitted assignment satisfies the formula. A graph-coloring success receives credit only when the emitted coloring is valid. This evidence is stronger than a category label, but it remains asymmetric because it does not require an UNSAT or non-colorability proof.

For SAT, the pair-level criterion can be written as

$$\text{ADR}^{+w} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[f(x_i^+) = 1 \wedge \text{ValidWitness}(x_i^+) \wedge f(x_i^-) = 0].$$

The graph-coloring criterion ADR^{+a} replaces the satisfying assignment with a valid coloring. A credited pair therefore demonstrates more than agreement with two labels: the model must also expose a constructive object that satisfies the task semantics on the positive instance. Within the available archives, this makes W the closest operational proxy to reasoning ability that is directly machine-checkable. It remains a proxy rather than ground truth because witness emission depends on elicitation and formatting, and because the negative arm lacks a symmetric proof requirement.

For any reported metric M , we use

$$\Delta_M^W = M - W$$

as a descriptive metric–witness gap. A positive value is the amount of observable credit assigned by M that is not accompanied by witness-grounded success under the same archived slice. It is not an estimator of unobserved internal reasoning.

The refreshed 2-SAT archive provides an additional two-sided follow-up, ADR^{+wu} . A pair receives credit only when the SAT response contains a verified satisfying assignment and the UNSAT follow-up contains a solver-verified unsatisfiable clause subset. The source table computes this quantity only on pairs for which both follow-up artifacts are available and uses “–” when no pair is eligible. Consequently, ADR^{+wu} is an audit of the complete follow-up protocol, not an unconditional lower bound on ADR^{+w} , and it is not used in the cross-model ordering statistic below.

Shortcut-adjusted margins. The shortcut-margin archive uses its own internally consistent factorized lower bound, denoted $\underline{s}_M^{\text{CP}}$, and reports

$$\mathcal{M}_0 = \underline{s}_M^{\text{CP}} - C_0, \quad \mathcal{M}_{\text{sc}} = \underline{s}_M^{\text{CP}} - \widehat{C}_{\text{sc}}, \quad C_0 = 0.25.$$

For CNF tasks, the specified shortcut feature map contains mean, minimum, and maximum clause length, literal-polarity ratio, and variable coverage. The graph-coloring implementation uses analogous degree and density statistics. Within each matched group, the archive computes the two-sample KS distance for each feature and takes the maximum one-dimensional threshold separation, $\hat{d}_G^{\text{obs}}$. To correct the upward bias caused by maximizing over features, it permutes the positive/negative labels within the group, recomputes the max-KS statistic, and subtracts the 0.95 quantile of the permutation reference distribution. The reported experiments use 200 permutations by default:

$$\hat{d}_{\text{short}}(G) = \max \left\{ 0, \hat{d}_G^{\text{obs}} - Q_{0.95} \left(\hat{d}_G^{\text{perm}} \right) \right\}, \quad \widehat{C}_{\text{sc}}(G) = \left(\frac{1 + \hat{d}_{\text{short}}(G)}{2} \right)^2.$$

The reported $\widehat{C}_{\text{sc}}$ is macro-averaged over evaluable groups in the model-size slice. These are design-conditional margins. The adversary battery contains single-feature threshold rules rather than all possible shortcuts, and the reported residual ceiling uses a permutation-calibrated point estimate rather than an upper confidence bound for shortcut distinguishability. Accordingly, a negative $\mathcal{M}_{\text{sc}}$ remains negative under a stricter ceiling, whereas a positive value could shrink if uncertainty in the shortcut estimate were also charged.

5.5 Derived Analyses

Strict model-ordering discordance. For each domain and problem size, we compare every model pair for which both the candidate metric and W differ at the three-decimal precision reported by the source archive. For metric M ,

$$D_M = \frac{\sum_N \sum_{a < b} \mathbf{1} \left[(M_{a,N} - M_{b,N})(W_{a,N} - W_{b,N}) < 0 \right]}{\sum_N \sum_{a < b} \mathbf{1} \left[M_{a,N} \neq M_{b,N} \wedge W_{a,N} \neq W_{b,N} \right]}.$$

Comparisons are made only within the same task and size. Ties are excluded rather than broken using unreported precision. The statistic does not assume that witness success completely measures reasoning; it asks whether a metric induces the same model ordering as a stronger observable criterion.

Pair-construction sensitivity. The matched-separability tables report ADR under the original pairing and under maximum and minimum feasible rematchings of the same positive and negative predictions. We define

$$\Delta_{\text{match}} = \text{ADR}_{\text{max}} - \text{ADR}_{\text{min}}.$$

A nonzero Δ_{match} means that the identical marginal prediction record supports different matched scores solely because the pairing rule changes.

Threshold and shortcut-account sensitivity. For the matched-separability archive, a prefix boundary is

$$N^*(\tau) = \max \left\{ N : \underline{s}^{\text{CP}}(n) \geq \tau \text{ for every reported } n \leq N \right\}.$$

The prefix convention prevents an isolated recovery at a larger size from being interpreted as sustained behavior. In the shortcut-margin archive, a boundary is the largest tested size at which the corresponding CP margin remains positive. We compare the ideal-floor boundary from $\mathcal{M}_0$ with the residual-ceiling boundary from $\mathcal{M}_{\text{sc}}$.

Table 2. Representative rows from the matched-separability archive. W denotes ADR^{+w} for SAT and ADR^{+a} for graph coloring. $\Delta_M^W = M - W$ makes the distance from the witness-grounded proxy explicit. Positive gaps are output-level credit without corresponding witness-validated pair success.

Domain	Model	N	Acc	ADR	$\hat{s}$	$\underline{s}^{\text{CP}}$	$\underline{s}^W$	W	Δ_{Acc}^W	Δ_{ADR}^W	$\Delta_{\underline{s}^{\text{CP}}}^W$
3-SAT	GPT-5	25	0.864	0.736	0.732	0.612	0.615	0.673	+0.191	+0.063	−0.061
3-SAT	Opus 4.7 (med. think)	25	0.935	0.759	0.783	0.592	0.602	0.655	+0.280	+0.104	−0.063
3-SAT	GPT-5	50	0.573	0.173	0.168	0.099	0.105	0.064	+0.509	+0.109	+0.035
Graph coloring	GPT-OSS-20B (think)	10	0.886	0.818	0.781	0.710	0.700	0.818	+0.068	0.000	−0.108
Graph coloring	Llama-Nemo-49B (r-on)	10	0.909	0.818	0.818	0.751	0.740	0.227	+0.682	+0.591	+0.524
Graph coloring	Nemo-H-47B (r-on)	8	0.668	0.338	0.346	0.293	0.288	0.002	+0.666	+0.336	+0.291

5.6 Analysis Protocol and Claim Discipline

The matched and shortcut-aware secondary calculations use the three-decimal values printed in the frozen LaTeX exports. The group analysis reads the current raw prediction records and exports full-precision CSV rows separately for 3-SAT, 2-SAT, and graph coloring. We report exact row counts and descriptive contrasts rather than treating model-size rows as independent replications. The source CP and Wilson calculations remain the primary finite-sample summaries. The source protocol specifies a matched-pair cluster bootstrap, with at least 2,000 replicates recommended, to preserve within-pair and shared-template dependence. The supplied summary exports do not contain raw prediction records or bootstrap replicates, so we do not reconstruct or invent bootstrap intervals from rounded tables. Similarly, we retain “−” for unavailable 2-SAT two-sided witness follow-ups rather than treating them as zero. These limitations constrain the empirical claim to the archived observables and reported uncertainty procedures.

6 Constraint-Reasoning Results

6.1 RQ1: Metric Levels and Model Orderings Diverge from Witness Evidence

Conventional label metrics are usually far above the witness-grounded reasoning proxy. Accuracy exceeds W in 18/19 3-SAT rows, 55/59 2-SAT rows, and all 45/45 graph-coloring rows. Across the three domains, this is 118/123 rows (95.9%); the other five rows are ties at the source precision. ADR, despite requiring a correct pair of labels, exceeds W in 15/19, 47/59, and 40/45 rows, respectively. This is 102/123 rows (82.9%), with the remaining rows tied. The direction is therefore systematic rather than confined to a few examples. These differences are not metric errors. Accuracy measures marginal label agreement, ADR measures correct paired labels, and W additionally requires a constructive positive-side artifact. The invalid step would be to interpret the first two as interchangeable measurements of witness-grounded, demonstrated reasoning.

Table 2 illustrates both level gaps and ranking reversals. At 3-SAT $N = 25$, Opus 4.7 has higher accuracy, ADR, $\hat{s}$, G-mean, balanced accuracy, and MCC than GPT-5, yet its witness rate is lower, $0.655 < 0.673$. The CP lower bound happens to restore the witness ordering for this pair, $0.592 < 0.612$, but this is not a universal property. At graph-coloring $N = 10$, Llama-Nemo-49B has higher accuracy, $\hat{s}$, and $\underline{s}^{\text{CP}}$ than GPT-OSS-20B while producing far fewer valid colorings, 0.227 versus 0.818. Nemo-H-47B at $N = 8$ gives a sharper example: accuracy is 0.668 and $\underline{s}^{\text{CP}} = 0.293$, but the valid coloring rate is 0.002.

The gap columns show why the distinction matters in absolute level, not only in rank. Llama-Nemo-49B at graph-coloring $N = 10$ receives 0.909 accuracy and 0.818 ADR, but only 0.227 of pairs include a valid positive-side coloring. The corresponding gaps are 0.682 and 0.591. Nemo-H-47B at $N = 8$ has a witness rate of 0.002, leaving gaps of 0.666 for accuracy and 0.336 even for ADR.

Table 3. Strict model-ordering discordance with the positive-side witness reference, reported as discordant comparisons divided by all strict same-domain, same-size comparisons. Ties at the three-decimal source precision are excluded.

Domain	Acc	F1 ₊	F1 _−	ADR	$\hat{s}$	$\underline{s}^{\text{CP}}$	G-mean	BA	MCC
3-SAT	3/16	4/16	2/16	4/16	4/16	3/16	4/16	3/16	3/16
2-SAT	10/115	25/116	10/111	11/116	12/116	9/116	13/116	12/116	14/116
Graph coloring	25/62	34/62	21/63	24/62	26/63	27/63	30/63	24/62	25/62

Table 4. Representative metric–witness gaps in the shortcut-margin archive. The last margin subtracts the permutation-calibrated structural shortcut ceiling from the factorized CP lower bound. It is reported as a design-conditional diagnostic, not as a standalone reasoning metric.

Model	N	Acc	F1 ₊	F1 _−	ADR ⁺ _w	Shortcut-adjusted CP margin
GPT-5	40	0.645	0.466	0.735	0.164	−0.196
DeepSeek-Reasoner	25	0.626	0.454	0.716	0.144	−0.189

A confidence bound does not necessarily close the construct gap: $\underline{s}^{\text{CP}} - W = 0.524$ and 0.291 in these two rows. Negative entries in the last column have a different meaning. They show that a conservative two-sided lower bound can fall below observed witness production; they do not make the bound a better proxy for constructive reasoning.

The disagreement is not limited to selected examples. Table 3 reports strict pairwise ordering discordance. On graph coloring, accuracy reverses the witness ordering in 25/62 strict comparisons, ADR in 24/62, $\hat{s}$ in 26/63, and even $\underline{s}^{\text{CP}}$ in 27/63. Every reported metric has a nonzero discordance count on every task. Thus, strengthening the evaluation from accuracy to matched or confidence-bounded separability changes the observable criterion without making its model ordering monotone in witness-validated success. This is an empirical instance of Proposition 2, not a claim that the witness ordering is itself a complete ordering of reasoning ability.

The shortcut-margin archive independently shows that class-balanced metrics can remain non-trivial after a stricter, design-conditional margin has collapsed. Table 4 gives two direct examples. GPT-5 at $N = 40$ retains accuracy 0.645 and negative-class F1 0.735, but $\text{ADR}^{+w} = 0.164$. DeepSeek-Reasoner at $N = 25$ has accuracy 0.626 and negative-class F1 0.716, but $\text{ADR}^{+w} = 0.144$. The accuracy–witness gaps are 0.481 and 0.482, respectively, while both shortcut-adjusted CP margins are negative.

For GPT-5 at 3-SAT $N = 30$, accuracy is 0.735, F1_− is 0.787, MCC is 0.533, G-mean is 0.694, balanced accuracy is 0.734, ADR is 0.486, and $\text{ADR}^{+w} = 0.440$. The CP margin above the ideal 0.25 floor is still +0.117, but after charging the specified residual shortcut ceiling the CP margin is −0.023. At $N = 50$, F1_− remains 0.695 and balanced accuracy remains 0.573, while $\text{ADR}^{+w} = 0.064$, $M_0(\text{CP}) = -0.151$, and $M_{\text{sc}}(\text{CP}) = -0.240$. A corresponding 2-SAT example is gpt-oss-20b at $N = 20$: accuracy is 0.742, F1_− is 0.785, MCC is 0.528, and $\text{ADR}^{+w} = 0.424$, whereas the ideal CP margin is only +0.040 and the residual CP margin is −0.202. The result is not that the residual margin measures reasoning correctly; it is that reasonable observable criteria support materially different claims on the same archived outputs.

Per-size shortcut-margin trajectories. Figure 1 reports the full per-size trajectories. Its dashed curves are existing metrics: the label-only summaries accuracy, per-class F1, and ADR, together with

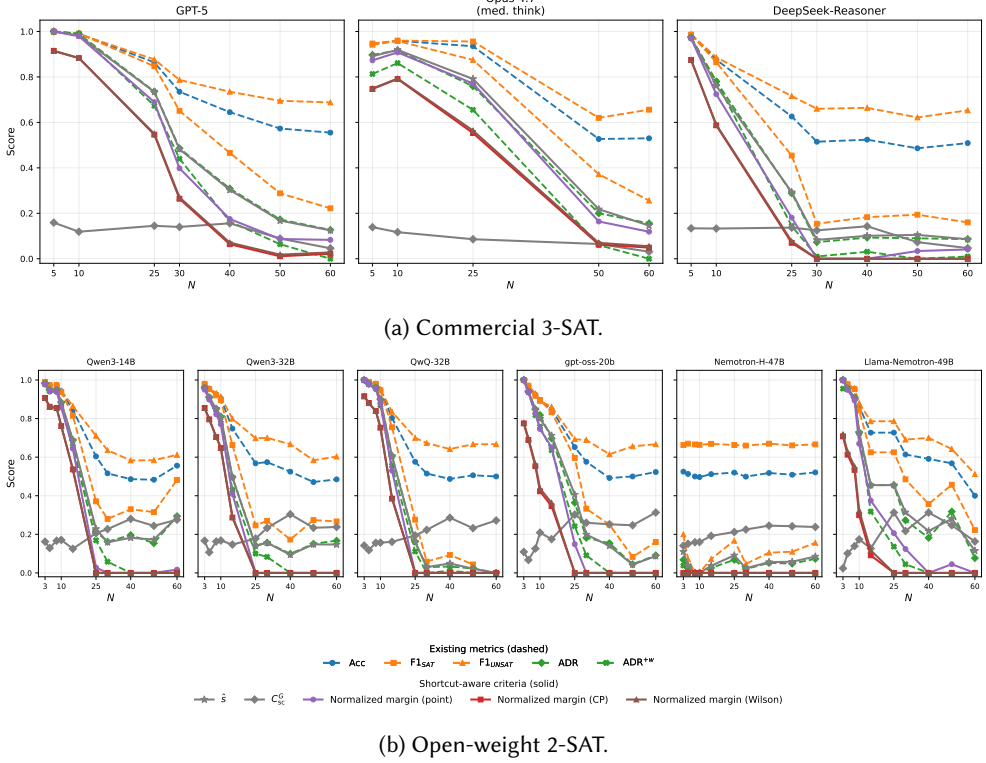


Fig. 1. Shortcut-aware trajectories under the permutation-calibrated residual shortcut ceiling. Conventional metrics often remain visibly nonzero after the witness-grounded and confidence-bounded criteria have fallen sharply. Under the No-Metrics interpretation, the separation is a claim gap: label agreement persists after evidence of demonstrated constructive reasoning has weakened, and the residual margin makes an additional design-conditional shortcut charge.

the witness-grounded ADR^{+w} ; its solid curves report two-sided separability, the charged shortcut ceiling, and the normalized shortcut-adjusted margins. Figure 2 holds the frozen model outputs fixed and instead plots only residualized-minus-original differences for quantities affected by the nuisance specification. The first figure asks how much apparent performance survives stronger evidence requirements. The second asks how much the shortcut-adjusted conclusion moves when the shortcut features are residualized against raw complexity covariates.

Together, the figures reinforce two conclusions from Table 2. First, accuracy and per-class F1 can remain moderate when ADR^{+w} , the closest machine-checkable proxy to demonstrated reasoning in these archives, is small or zero. Second, a shortcut-adjusted confidence margin answers a different, specification-bound question: whether matched category behavior clears a stated nuisance ceiling. Complexity residualization changes that ceiling and the resulting margins without changing predictions or witnesses. The margin therefore neither supersedes the witness nor identifies the model's internal mechanism.

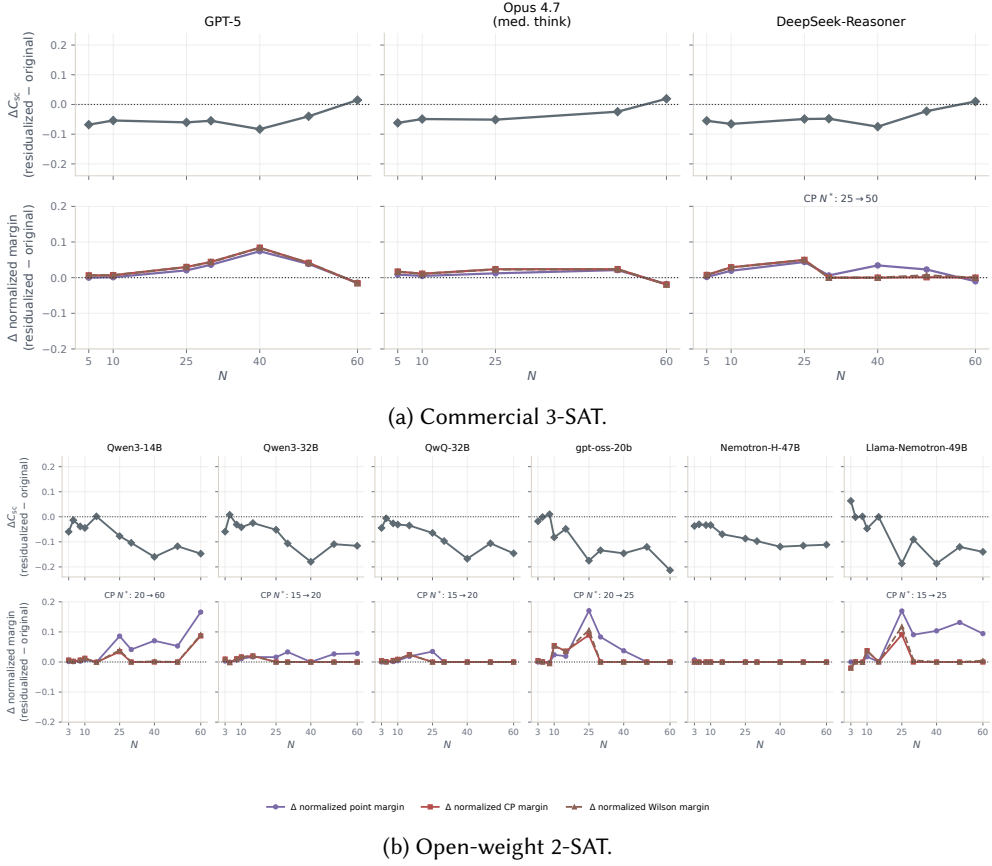


Fig. 2. Sensitivity of shortcut-aware conclusions to complexity residualization. Each trajectory is the complexity-residualized specification minus the original permutation-calibrated specification for the same frozen model outputs. The upper row reports ΔC_{sc} ; the lower row reports changes in normalized point, Clopper–Pearson, and Wilson margins. Values above or below zero indicate that residualization respectively increases or decreases the plotted quantity. Conventional metrics and witness-grounded quantities are omitted because they are unchanged. $CP N^*$ annotations use the largest source-grid N with a strictly positive full-precision signed CP margin; $N = 20$ is omitted from the 2-SAT trajectories to match Fig. 1. The figure isolates nuisance-model sensitivity; it neither selects a universally correct shortcut specification nor measures a change in latent reasoning ability.

6.2 RQ2: Two-Sided Criteria Separate One-Sided, Label-Only, and Witness-Level Success

Matched two-sided summaries expose behavior that marginal metrics can obscure. GPT-5 at 3-SAT $N = 60$ has positive-arm recall 0.127 and negative-arm recall 0.982. Accuracy remains 0.555 and $F1_-$ remains 0.688, but $\hat{s} = 0.125$, $\underline{s}^{CP} = 0.067$, and $ADR^{+w} = 0$. The arm decomposition localizes the failure: the model retains near-complete recognition of one category while losing the other. A large per-class score on the easy arm cannot compensate in the product criterion.

The converse separation is equally important. A model can retain two-sided *label* discrimination without constructing a valid positive witness. The graph-coloring rows in Table 2 show this directly. Llama-Nemo-49B at $N = 10$ has $\underline{s}^{CP} = 0.751$ but a valid-coloring rate of only 0.227; Nemo-H-47B

Table 5. Representative 2-SAT rows with the two-sided witness follow-up. ADR^{+wu} requires a verified SAT assignment and a verified UNSAT clause subset and is computed only on follow-up-eligible pairs; it is therefore reported as a separate audit rather than treated as an unconditional lower bound on ADR^{+w} . $\text{ADR}_{\text{all}}^{+wu}$ applies the same two-sided requirement but uses every complete pair at that N as the denominator, so pairs without a follow-up record count as failures; it is computed by the group-analysis code on the current prediction files (Section 5.2), whereas the other columns are archived values.

Model	N	ADR	ADR^{+w}	ADR^{+wu}	$\underline{s}^{\text{CP}}$	$\text{ADR}_{\text{all}}^{+wu}$
gpt-oss-20b (reason)	3	1.000	1.000	0.818	0.976	0.273
Qwen3-14B (think)	3	0.981	0.981	0.118	0.950	0.039
Qwen3-32B (think)	10	0.810	0.777	0.281	0.753	0.066
gpt-oss-20b (reason)	10	0.727	0.682	0.333	0.631	0.061
gpt-oss-20b (reason)	15	0.545	0.500	0.000	0.485	0.000
QwQ-32B (think)	20	0.283	0.242	0.231	0.222	0.030

at $N = 8$ has $\underline{s}^{\text{CP}} = 0.293$ but a valid-coloring rate of 0.002. The lower bound therefore supports a claim about matched label recognition under the constructed groups, not a claim about constructive graph reasoning. In these rows, ADR^{+a} is the more direct proxy for demonstrated reasoning because the predicted coloring itself is checked against the graph constraints. The large $\underline{s}^{\text{CP}} - \text{ADR}^{+a}$ gaps show that two-sided category recognition can remain far from that constructive standard.

The 2-SAT follow-up strengthens this decomposition by adding evidence on the UNSAT side. Table 5 compares ordinary ADR, positive-side witness ADR, and the follow-up-eligible two-sided witness audit. Near-perfect label and SAT-witness results do not imply reliable production of both artifacts. At $N = 3$, Qwen3-14B has $\text{ADR} = 0.981$ and $\text{ADR}^{+w} = 0.981$, but $\text{ADR}^{+wu} = 0.118$. At $N = 10$, Qwen3-32B has $\text{ADR} = 0.810$, $\text{ADR}^{+w} = 0.777$, and $\text{ADR}^{+wu} = 0.281$. For gpt-oss-20b, the two-sided audit falls from 0.333 at $N = 10$ to 0 at $N = 15$, even though the label-only ADR remains 0.545 and the positive-side witness rate remains 0.500.

This follow-up does not by itself identify internal reasoning. A failure can reflect inability to construct the requested artifact, an elicitation failure, a parsing failure, or failure of the underlying reasoning process. Its value is measurement separation: label discrimination, positive construction, and available two-sided construction are empirically different observables and should not be collapsed into one score. Among them, the verified two-sided artifact is the strongest evidence that the model successfully externalized the intended reasoning on that pair, although it remains an elicitation- and protocol-dependent lower-bound-like proxy rather than a complete measure of latent ability.

The choice between Clopper–Pearson and Wilson intervals is not the main source of these gaps. Across the 19, 59, and 45 constraint rows, the mean absolute difference $|\underline{s}^{\text{CP}} - \underline{s}^{\text{W}}|$ is 0.0043, 0.0050, and 0.0040, respectively; the maximum is 0.011 in every domain. Agreement between the interval formulas strengthens the conclusion that a low two-sided lower bound is not produced by one particular binomial approximation. It does not identify the mechanism behind a high lower bound.

6.3 RQ3: Controlled Conclusions Depend on Construction and Calibration

Pair matching. Pair rematching provides a direct test of construction dependence. The mean Δ_{match} is 0.026 on 3-SAT, 0.023 on 2-SAT, and 0.072 on graph coloring. The maximum spreads are 0.080, 0.182, and 0.364. For gpt-oss-20b at 2-SAT $N = 25$, the same predictions yield ADR values from 0.182 to 0.364 under admissible rematchings. For QwQ-32B at graph coloring $N = 40$, the

Table 6. Overall group-level results from the current raw predictions. $\text{SGADR} = \text{SGADR}_1$ and $\text{SGADR}_\star = \text{SGADR}_{\star,1}$. $A_{\geq k}$ is the fraction of evaluable groups containing at least k successful matched pairs. GADR , ADR , SGADR , $\text{SGADR}_\star$, and $A_{\geq k}$ are label-only. W is the witness-aware pair rate on the same pairs with the same denominator (ADR^{+w} for SAT, ADR^{+a} for graph coloring): a pair counts only if both labels are correct and the positive-side response carries a satisfying assignment or proper coloring verified by the domain’s established validator. $A_{\geq k}^W$ is the fraction of evaluable groups with at least k witness-verified successful pairs. †: in these 2-SAT rows the current prediction files contain many responses without a final answer object (recorded as parse failures); a label is recovered from the free text, but no assignment exists to verify, so W and $A_{\geq k}^W$ are lower bounds relative to the archived witness values used in RQ1 (Section 5.2).

Task	Model	GADR	ADR	W	SGADR	SGADR $_\star$	$A_{\geq 1}$	$A_{\geq 5}$	$A_{\geq 10}$	$A_{\geq 1}^W$	$A_{\geq 10}^W$
3-SAT	GPT-5	0.539	0.526	0.452	0.421	0.276	0.942	0.507	0.203	0.768	0.188
	Opus 4.7	0.600	0.644	0.544	0.514	0.415	0.878	0.449	0.224	0.633	0.184
	DeepSeek-Reasoner	0.342	0.350	0.289	0.249	0.123	0.771	0.286	0.171	0.486	0.171
2-SAT	Llama-Nemotron-Super-49B †	0.556	0.558	0.039	0.395	0.311	1.000	0.545	0.273	0.364	0.000
	Nemotron-H-47B	0.038	0.038	0.004	0.017	0.001	0.147	0.029	0.001	0.010	0.000
	gpt-oss-20b	0.540	0.551	0.485	0.423	0.303	0.903	0.613	0.258	0.742	0.258
	QwQ-32B †	0.460	0.464	0.218	0.397	0.215	0.701	0.485	0.340	0.598	0.010
	Qwen3-14B †	0.535	0.535	0.328	0.422	0.286	0.921	0.556	0.338	0.662	0.073
	Qwen3-32B †	0.453	0.455	0.235	0.338	0.207	0.884	0.455	0.248	0.653	0.008
Graph coloring	Llama-Nemotron-Super-49B	0.318	0.318	0.062	0.192	0.101	0.839	0.258	0.065	0.258	0.000
	Nemotron-H-47B	0.134	0.134	0.001	0.063	0.018	0.424	0.132	0.010	0.005	0.000
	QwQ-32B	0.423	0.424	0.182	0.285	0.179	0.929	0.439	0.133	0.643	0.000
	Qwen3-14B	0.320	0.320	0.165	0.184	0.102	0.797	0.349	0.050	0.456	0.000
	Qwen3-32B	0.313	0.314	0.172	0.162	0.099	0.856	0.341	0.015	0.553	0.000
	gpt-oss-20b	0.292	0.316	0.280	0.185	0.100	0.690	0.241	0.034	0.483	0.034

range is 0.155 to 0.436. For Llama-Nemo-49B at graph coloring $N = 60$, the original ADR and minimum ADR are 0, whereas the maximum rematching reaches 0.364. The model outputs are unchanged; only the construction of the controlled unit differs.

Group aggregation and repeated-success thresholds. We apply the metrics in Section 4.3 to the current raw predictions for all three constraint domains. The analysis covers three commercial 3-SAT configurations and six open-weight configurations for each of 2-SAT and graph coloring. Table 6 reports the overall rows at $\lambda = 1$; the underlying analysis also exports every model–size row, the full lambda sweep, group-level audit rows, completion counts, and thresholds from one through ten successful pairs. The same aggregation code also computes witness-aware counterparts (W and $A_{\geq k}^W$) by calling each domain’s established witness validator on the positive-side response before crediting a pair, so the label-only and witness-aware columns share pairs, groups, and denominators.

Macro and micro weighting rarely drives the main conclusion in these data, but it is not interchangeable. Across model–size slices, the mean absolute gap $|\text{GADR} - \text{ADR}|$ is 0.0077 on 3-SAT, 0.0028 on 2-SAT, and 0.0014 on graph coloring. The largest gaps are 0.081, 0.142, and 0.017, respectively. Thus, equal-group and equal-pair weighting usually agree because most groups contribute similar numbers of complete pairs, but a single aggregate can still conceal a substantial weighting decision in individual slices.

The nonlinear summaries reveal cross-group heterogeneity that pooled ADR does not show. For Opus 4.7 on 3-SAT, GADR is 0.600, while $\text{SGADR}_1 = 0.514$ and $\text{SGADR}_{\star,1} = 0.415$. For QwQ-32B on graph coloring, the same sequence is 0.423, 0.285, and 0.179. QwQ-32B on 2-SAT has the largest overall between-group variance, 0.185. These differences indicate that moderate pair-level performance can be concentrated in some controlled neighborhoods rather than sustained

uniformly. The lambda sweep preserves the identities in Equation (4); changing λ changes the variance charge, not the underlying prediction record.

The repeated-success results provide the clearest test of stable behavior. The largest overall $A_{\geq 10}$ is 0.340 at the precision shown in Table 6, even though $A_{\geq 1}$ is as high as 1.000. Across model-size slices, $A_{\geq 10} = 0$ in 12/19 3-SAT rows, 43/66 2-SAT rows, and 33/45 graph-coloring rows. Qwen3-32B on graph coloring illustrates the gap: ADR is 0.314, and 85.6% of evaluable groups contain at least one successful pair, but only 1.5% contain ten. Qwen3-14B on 2-SAT has GADR and ADR of 0.535, whereas $A_{\geq 10} = 0.338$. Opus 4.7 reaches ADR = 0.644 on 3-SAT, but only 22.4% of groups contain ten successful pairs. Correct labels therefore occur much more often than persistent success across variants of the same controlled construction.

This drop is consistent with the invariance motivation for group evaluation. If a model had formed a stable representation of the decision-relevant structure, incidental changes within the controlled neighborhood should not turn most groups from success to failure. The observed threshold profiles do not establish the opposite causal claim, because difficulty, correlated errors, and missing outputs can also break consistency. Completion is high for most overall rows, but ranges from 0.711 to nearly 1; an absolute count threshold must therefore be reported with completion. The supported conclusion is behavioral: isolated or sparse pair success overstates how often the same model succeeds repeatedly within a group.

Group metrics versus witness evidence. Group consistency does not close the larger gap to witness-grounded evidence. GADR, SGADR, and $A_{\geq k}$ reuse the same label-pair outcomes as ADR; none checks whether a SAT label is accompanied by a satisfying assignment or whether a colorable label is accompanied by a valid coloring. The witness-aware columns of Table 6 apply the same aggregation to the same groups after that check, so the distance between the label-only and witness-aware columns is the part of group-level consistency that was never witnessed. Four observations follow. Unless stated otherwise, the numbers below use the 130 model-size slices; the 3-SAT rows and the non-dagger rows are the ones whose current inputs reproduce the archived witness summaries exactly.

Level. W is below ADR in every overall row and every slice. Across slices the ADR- W gap has mean 0.171 and median 0.102. Even in the best-behaved rows the witness-verified share of label-pair successes is 0.83–0.89 (GPT-5 0.86, Opus 4.7 0.85, DeepSeek-Reasoner 0.83, gpt-oss-20b 0.88 on 2-SAT and 0.89 on graph coloring); in the other graph-coloring rows it is 0.00–0.55, and Nemotron-H-47B retains 0.10 on 2-SAT. Group weighting does not help: GADR- W is within 0.03 of ADR- W in 14/15 rows, because GADR and ADR agree on which pairs count and disagree only on how groups are weighted.

Nonlinearity is not a witness check. SGADR and SGADR $_{\star}$ are numerically closer to W than ADR is in 105/130 and 102/130 slices, and on 3-SAT all three overall SGADR values fall slightly below W (−0.03 to −0.04). This proximity is coincidental. SGADR subtracts a within-group variance term, whereas W removes pairs without a verified artifact; the two corrections respond to different properties of the outputs. Accordingly, SGADR lands below W in 40/130 slices, yet exceeds W by more than 0.10 in five overall rows (Llama-Nemotron-Super-49B by 0.36 and QwQ-32B by 0.18 on 2-SAT; Llama-Nemotron-Super-49B by 0.13, QwQ-32B and Qwen3-32B by 0.10 elsewhere). A penalty for inconsistency cannot be read as a proxy for demonstrated construction.

Ordering. Ranking the six open-weight models by any label-only group metric agrees with the ranking by W on only 9–10 of the 15 model pairs on graph coloring, and on 9–13 of 15 on 2-SAT. The repeated-success statistic is the least aligned: $A_{\geq 10}$ agrees with W on 9/15 pairs in both domains. On 2-SAT, QwQ-32B has the largest $A_{\geq 10}$ (0.340) but $A_{\geq 10}^W = 0.010$, whereas gpt-oss-20b retains its

Table 7. Matched-separability prefix boundaries under alternative reporting thresholds. Boundaries are comparable only within the same task family.

Domain	Model	$N^*(0.50)$	$N^*(0.25)$	$N^*(0.10)$	$N^*(0.05)$
3-SAT	GPT-5	25	30	40	60
3-SAT	Opus 4.7 (med. think)	25	25	50	60
3-SAT	DeepSeek-Reasoner	10	10	25	25
2-SAT	Qwen3-14B (think)	15	20	50	50
2-SAT	Qwen3-32B (think)	10	15	20	30
2-SAT	QwQ-32B (think)	10	15	20	25

$A_{\geq 10} = 0.258$ unchanged under the witness requirement. Consistent label success across a group therefore neither implies nor orders consistent witnessed success.

Thresholds. The repeated-success rates degrade fastest. $A_{\geq 10}^W$ is at most 0.258 and is 0.000 in five of the six graph-coloring rows; $A_{\geq 10} > 0$ in 42 slices, and in 23 of them $A_{\geq 10}^W = 0$. Sixty-five slices have $A_{\geq 1} = 1.000$, yet their $A_{\geq 1}^W$ ranges down to 0.000. Fifty-two slices have $W = 0$; in 50 of them ADR, SGADR, and $A_{\geq 1}$ are all positive, including 37 slices outside the dagger rows (GPT-5 at 3-SAT $N = 60$: ADR 0.127, $A_{\geq 1} = 0.700$, $W = 0$). On 3-SAT, where the archive is reproduced exactly, GPT-5 has $A_{\geq 1} = 0.942$ but $A_{\geq 1}^W = 0.768$, and Opus 4.7 falls from $A_{\geq 10} = 0.224$ to $A_{\geq 10}^W = 0.184$.

In the source-separated witness analysis of RQ1, ADR already exceeds the positive-side witness proxy in 15/19, 47/59, and 40/45 rows for 3-SAT, 2-SAT, and graph coloring. The group analysis sharpens that finding: applying a group transformation to label outcomes cannot supply the missing witness, and neither equal-group weighting, a variance penalty, nor a repeated-success threshold moves a label-only summary reliably toward W . Group metrics measure persistence of label agreement within a controlled construction; they remain summaries of predictions, not evidence of the constructive ability that W checks. The stronger operational target is therefore conjunctive: correct differentiation, persistence across controlled variants, and a checkable witness where one exists. The current group metrics measure the second property more directly than ordinary ADR, while ADR^{+w} and ADR^{+a} remain closer to demonstrated reasoning because they verify a semantic artifact. Neither alone identifies latent reasoning.

Reporting threshold. Table 7 reports the matched-separability prefix boundaries under four thresholds. GPT-5’s 3-SAT boundary is 25, 30, 40, or 60 when τ is 0.50, 0.25, 0.10, or 0.05. Qwen3-14B’s 2-SAT boundary changes from 15 to 20, 50, and 50 under the same sequence. These values answer different effect-size questions; they are not competing estimates of a threshold-free latent capability.

Shortcut account and confidence requirement. Table 8 summarizes the shortcut-margin archive. The ideal and residual views use the same model outputs and lower-bound protocol; they differ only in the charged ceiling. GPT-5’s 3-SAT boundary changes from 30 under $\mathcal{M}_0$ to 25 under $\mathcal{M}_{sc}$. On 2-SAT, Qwen3-32B changes from 15 to 10, gpt-oss-20b from 20 to 15, and Llama-Nemotron-Super-49B from 10 to 8. Other SAT rows are unchanged because the added shortcut charge does not cross a tested boundary. We exclude the graph-coloring shortcut-ceiling contrast from this table: the frozen runner invokes the same complexity-residualized configuration for the files labeled as its raw and residualized branches, so those exports cannot support a between-branch comparison.

Table 8. SAT shortcut-margin boundaries. N^* is the largest tested size with a positive CP margin under the ideal 0.25 floor or the permutation-calibrated residual shortcut ceiling. A dash denotes no positive boundary in the reported grid. These are source-archive, design-conditional boundaries rather than mechanism-independent capability estimates.

Domain	Model	$\mathcal{M}_0 N^*$	$\mathcal{M}_{sc} N^*$
3-SAT	GPT-5	30	25
3-SAT	Opus 4.7 (med. think)	25	25
3-SAT	DeepSeek-Reasoner	10	10
2-SAT	Qwen3-32B (think)	15	10
2-SAT	gpt-oss-20b (reason)	20	15
2-SAT	Nemotron-H-47B (r-on)	–	–
2-SAT	Llama-Nemotron-Super-49B (r-on)	10	8

The matched-separability and witness results for graph coloring remain usable because they come from the separate archive and do not depend on that contrast.

The lower-bound requirement can change the conclusion even before the shortcut account changes. GPT-5 at 3-SAT $N = 40$ has an ideal point margin of +0.053, but its CP margin is -0.040 . gpt-oss-20b at 2-SAT $N = 25$ has an ideal point margin of +0.156, but its CP margin is -0.045 . Thus a point estimate above a floor is not equivalent to a lower-bound claim. The residual view adds another conditional choice: which feature battery is admitted as a plausible shortcut class. Because the reported $\mathcal{M}_{sc}$ ceiling is itself a calibrated point estimate, its positive margins are best read as sensitivity results relative to the tested battery, not as universal shortcut exclusion.

Together, RQ3 shows that controlled metrics correct specific weaknesses while remaining design dependent. They prevent one arm from compensating for the other, preserve matched structure, can require repeated pair success, and can charge measured nuisance factors. Their numerical values and boundaries still depend on the pairing rule, equal-group versus equal-pair weighting, the order of nonlinear aggregation, the success-count threshold and effective group size, the minimum reported effect size, the uncertainty construction, and the stated adversary class. These dependencies are part of the evidence and must be reported rather than hidden behind a single scalar.

7 Discussion

What the experiments establish. The experiments provide distinct empirical counterparts to the three formal limitations. The witness-ordering reversals instantiate metric non-monotonicity: better output performance under one criterion can accompany worse performance under a stronger observable criterion. Pair-rematching, threshold, and shortcut-ceiling sensitivity instantiate the design dependence behind non-separability. None of these analyses can empirically distinguish two models that produce exactly the same outputs, which is precisely the mechanism non-identifiability of Proposition 1.

What the experiments do not establish. The witness reference is asymmetric and incomplete, and controlled metrics do not recover the latent pair (δ_U, ϵ_U) . Accordingly, we do not interpret a lower $\underline{s}^{CP}$ as proof of weaker internal reasoning or a higher witness rate as a complete reasoning score. The supported claims are conditional: better label agreement, stronger two-sided matched behavior, more constructive positive-side success, or greater robustness under a stated intervention.

A claim-matched reporting standard. A reasoning-oriented evaluation should report at least four layers of evidence. First, conventional performance metrics describe deployment-facing agreement. Second, arm-specific and controlled metrics expose one-sided and local-contrast failures. Third, checkable witnesses or stronger task-specific oracles validate outputs when available. Fourth, construction sensitivity, nuisance-factor perturbations, and dependence-aware uncertainty delimit the claim. A paper should state which layer supports each conclusion rather than collapsing all layers into a single leaderboard number.

Applicability across model architectures. Our conclusions do not depend on a particular model architecture. They apply to additive classifiers, attention-based models, graph neural networks, and other systems whenever evaluation reduces model behavior to predictions or scalar scores. These architectures differ in how they represent and combine information, but this difference does not change the interpretation of the resulting metric. A higher value establishes better performance under the metric’s observable criterion, not necessarily stronger reasoning. Section S6 of the supplementary material provides illustrative model-level instantiations of this argument.

Metrics as summaries of interventions. Controlled metrics are best interpreted as summaries of model behavior under particular interventions. Pair-level ADR summarizes whether predictions change correctly across matched opposite-label instances. Group and cluster metrics summarize whether this behavior remains consistent across related variants. Metrics over semantics-preserving perturbations summarize invariance to selected non-semantic changes. These measurements provide more targeted evidence than evaluation on independently sampled instances, but their interpretation depends on the quality of the interventions and the aggregation rule. A higher controlled-metric value therefore indicates stronger performance under the chosen intervention design, not necessarily stronger reasoning ability.

The group results make this distinction observable (Table 6). In 88/130 model-size slices, no group contains ten successful matched pairs, although many of the same slices contain isolated successes. Group evaluation therefore reveals instability that pooled ADR can average away. Yet a model can also be consistently right for a group-level non-semantic reason. Stable contrast behavior should consequently be read alongside witness validity, rather than as a replacement for it.

Intervention summaries become more informative when combined with task-specific validation. In SAT solving, a satisfying assignment can directly validate a positive answer. Controlled metrics summarize behavioral consistency, while witnesses test whether particular outputs satisfy stronger semantic requirements. Neither should be treated as a universal standalone measure of reasoning.

7.1 Beyond Classification and Ranking Metrics

Although our formal development focuses on classification, ranking, and controlled prediction metrics, the same underlying limitation applies more broadly to output-based evaluation metrics. Many ML/LLM-based software-engineering tasks are evaluated not by F1, MCC, or AUROC, but by task-specific observable outcomes. For example, code-reasoning benchmarks may use exact-match or pass@ k metrics, while test-generation systems may be evaluated by line or branch coverage. These metrics are appropriate for their tasks, but they still summarize observable success rather than the mechanism that produced it.

In a pass@ k benchmark, the evaluator observes whether at least one of the model’s generated outputs is accepted by a checker. This is stronger than unverified label agreement, because the output must satisfy a task-specific correctness condition. However, the metric still does not reveal whether the accepted output was produced by semantic reasoning, memorized patterns, benchmark-specific cues, or lucky sampling. Similarly, coverage-based test-generation metrics observe which lines or branches are executed by the generated tests. Coverage is useful evidence about structural

reachability, but it does not by itself show that the generator understood the program’s intended behavior, produced meaningful assertions, or exposed the semantic property of interest.

Thus, the limitation studied in this paper is not specific to a particular metric formula. It arises whenever the evaluator observes only a finite transcript of outputs and task-level success signals. Stronger checkers, witnesses, and coverage measurements can make the observed success event more meaningful, and they are often essential in practice. But unless the evaluation also tests semantic sensitivity, side-channel invariance, or mechanism-level evidence, output-based metrics alone cannot certify that success came from reasoning.

Cohen’s Kappa illustrates the same distinction. Kappa improves on raw accuracy by correcting observed agreement for agreement expected from the marginal class and prediction distributions. This makes it useful for detecting cases where high accuracy is partly explained by class prevalence or prediction bias. However, Kappa is still computed from the observable confusion matrix. It does not reveal whether the agreements counted in the confusion matrix were produced by semantic reasoning or by shortcut cues. Thus, Kappa can improve the statistical interpretation of benchmark agreement, but it does not solve the mechanism non-identifiability problem.

Broader use of prediction-output metrics. The concern studied in this paper is not unique to software-engineering benchmarks. Surveys of deep-learning evaluation show that many of the same families of metrics—accuracy, precision, recall, F1, ROC/AUC, error metrics, ranking metrics, and task-specific output metrics—are widely used across regression, classification, computer vision, natural-language processing, and retrieval-augmented generation [32]. This broader practice reinforces our point. These metrics are often useful summaries of observable performance, and in many settings they are the appropriate way to compare systems. Our claim is that when the intended construct is reasoning ability, an output metric does not by itself establish that the observed success was produced by semantic reasoning. The same accuracy, ranking, coverage, or pass/fail signal may be compatible with different underlying mechanisms. Software-engineering tasks make this issue especially salient because semantic evidence, side-channel cues, and benchmark artifacts can often be distinguished operationally through program transformations, repairs, witnesses, or perturbations.

8 Threats to Validity

Secondary-analysis scope and archive versioning. The study reanalyzes frozen summary exports rather than rerunning models. This prevents post-hoc prompt changes but limits us to the observables retained by the source tables. Some overlapping shortcut-margin and matched-separability rows differ slightly because the archives reflect different reruns or parsing revisions. We mitigate this threat by keeping the archives separate and by never combining row-level values across versions. A final artifact should replace source filenames with immutable commit hashes and include the raw per-instance predictions.

Witness asymmetry and elicitation. A satisfying assignment or valid coloring verifies constructive positive-side success, but the supplied experiments do not require machine-checkable UNSAT or non-colorability proofs. Witness failure can also arise from output format, parsing, or instruction following. We therefore use witness success as a stronger observable reference, not as a complete ground truth for reasoning.

Rounded values and post-hoc comparisons. The rank-discordance and sensitivity analyses use values rounded to three decimals in the frozen tables. We exclude ties and do not infer hidden precision, but a small number of orderings could change with full-precision values. The analyses are

also post-hoc. They should be replicated from per-instance outputs with a preregistered comparison protocol.

Dependence and pseudo-replication. Matched examples share seed families, templates, and prompts. Per-arm Clopper–Pearson and Wilson calculations use binomial assumptions that may be optimistic under intra-group dependence. The source matched-pair protocol specifies a matched-pair cluster bootstrap, but bootstrap replicates were not included in the supplied summary exports. The present manuscript therefore reports the available bounds and treats dependence-robust recomputation as required before a final empirical claim.

Controlled-construction quality. Minimal edits can introduce lexical, syntactic, or structural artifacts, and the shortcut feature set used by the residual ceiling is necessarily incomplete. Pair rematching exposes one part of construction sensitivity but does not enumerate every nuisance factor. A positive controlled result is conditional on the stated intervention and shortcut audit.

Model, prompt, and domain coverage. The constraint tasks cover several model families, but they do not establish universality across prompts, decoding settings, provider revisions, or constraint generators. Boundaries are comparable only within a task family and tested grid.

9 Related Work

9.1 Outcome-Based Evaluation of Reasoning-Oriented Tasks

A large fraction of reasoning-oriented evaluation is ultimately based on whether a model produces the correct final output. GSM8K, introduced by Cobbe et al. [5], evaluates grade-school mathematical word problems primarily through final-answer correctness; the same work trains verifiers on candidate solutions whose supervision is determined by whether the final numerical answer is correct. The MATH benchmark of Hendrycks et al. [14] raises the difficulty substantially by using competition mathematics problems and provides full step-by-step reference solutions, but its standard benchmark comparison is still based primarily on normalized final-answer accuracy. GPQA [25] similarly improves the quality of the evaluation task by using expert-authored graduate-level science questions that remain difficult for skilled non-experts even with web access. These benchmarks differ substantially in domain, difficulty, and data construction, but they exemplify a common evaluation pattern: a task is chosen because successful completion plausibly requires reasoning, while the reported score is determined mainly by the correctness of the final answer.

Such outcome metrics are valuable measures of task success. A higher GSM8K, MATH, or GPQA accuracy establishes that a system answers more benchmark items correctly under the evaluated protocol. It does not, however, identify the mechanism responsible for those correct answers. Correct outputs may be compatible with valid multi-step reasoning, memorized knowledge, retrieval, elimination strategies, shortcut cues, search-and-selection effects, or combinations of these mechanisms. Our work therefore does not dispute the validity of final-answer metrics for their immediate behavioral criterion. Instead, it studies the additional inferential step from better outcome performance to a claim of stronger reasoning ability.

9.2 Reasoning-Trace and Process Evaluation

A second line of work evaluates the generated reasoning process rather than only the final answer. ROSCOE [11] introduces a suite of automatic metrics for step-by-step rationales, including dimensions related to semantic alignment, semantic similarity, logical consistency, informativeness, factuality, and language quality. ReCEval [24] views a generated reasoning chain as an informal proof and evaluates two properties: correctness, including whether intermediate conclusions are supported and non-contradictory, and informativeness, including whether a step adds information

useful for deriving the generated answer. ReCEval implements these ideas using natural-language-inference models and $\mathcal{V}$ -information. These approaches provide strictly richer observable evidence than final-answer accuracy because they can distinguish reasoning traces that reach the same answer but differ in detectable intermediate errors.

Related work on process supervision makes the same distinction from the training perspective. Lightman et al. [19] compare outcome supervision, which labels a complete solution by its final result, with process supervision, which provides feedback on intermediate steps, and show that process supervision can substantially improve performance on MATH. This line of work demonstrates that intermediate-step evidence is useful and that final-answer supervision is not the only way to evaluate or train reasoning systems.

However, observable trace quality and mechanism faithfulness are distinct claims. Turpin et al. [33] show that chain-of-thought explanations can systematically rationalize answers influenced by hidden biasing features without reporting those influences. Lanham et al. [17] evaluate faithfulness by intervening on chain-of-thought traces and find substantial task- and model-dependent variation in how strongly answers depend on the stated reasoning. Thus, a high-quality rationale under ROSCOE, ReCEval, or a learned verifier is stronger evidence than final-answer correctness, but it does not by itself establish that the visible rationale is the causal mechanism that produced the answer.

This distinction is important for our analysis. Our formal results focus on prediction- and score-based metrics, while trace evaluators illustrate how richer observations can narrow, but not automatically eliminate, the gap between observable behavior and latent reasoning mechanisms.

9.3 Benchmark Reliability, Construct Validity, and Reasoning Claims

Recent work has increasingly treated reasoning evaluation as a measurement and validity problem rather than only a metric-selection problem. Salaudeen et al. [27] distinguish measurement instruments, measurements, evaluations, and claims, and argue that validity depends on whether the available evidence supports the particular claim being made. Their framework separates directly measurable criteria from broader latent constructs such as reasoning and emphasizes that increasingly broad capability claims require increasingly strong forms of validity evidence. Bean et al. [2] provide complementary empirical evidence through a systematic review of 445 LLM benchmarks. They analyze how target phenomena are operationalized through tasks and metrics and identify recurring construct-validity weaknesses in benchmark definition, sampling, scoring, and claim interpretation.

Most directly related to the empirical motivation of our work, Mousavi et al. [22] systematically audit SocialIQA, ToMi, and FauxPas-EAI and identify structural, semantic, and pragmatic problems in both benchmark items and evaluation protocols. Their re-evaluations show that scores can change materially after data cleaning, context changes, semantic rephrasing, and alternative scoring, without any corresponding change in the underlying model parameters. Their findings demonstrate that a benchmark score is a property of the broader evaluation pipeline, not a context-free measure of a model's reasoning ability, and that high scores may partly reflect format-specific regularities rather than stable inference from the intended information.

Our work is complementary to these validity-centered studies. Salaudeen et al. provide a general framework for asking what claims an evaluation can support, Bean et al. document how construct-validity problems arise in current benchmark practice, and Mousavi et al. show empirically how benchmark and protocol defects can distort reasoning scores. We study a narrower but more formal identification problem: even when benchmark labels and scoring procedures are taken as correct, finite prediction or score observations do not in general identify whether success was produced by semantic reasoning or by a non-semantic shortcut, and metric-based model

rankings need not be monotone in reasoning ability. In this sense, benchmark cleaning and better construct operationalization address important sources of invalidity, but do not by themselves resolve mechanism non-identifiability.

Prior surveys of classification metrics, such as Naidu et al. [23], address a different problem: selecting and standardizing statistical metrics for particular prediction settings. Our focus is not on identifying a universally preferable confusion-matrix or ranking metric. Rather, we ask what reasoning-oriented claims can be supported from the observable information retained by such metrics.

9.4 Controlled, Contrastive, and Intervention-Based Evaluation

A complementary strategy strengthens the evaluation design rather than only changing the scoring formula. Contrast sets perturb test instances in small but meaningful ways, often changing the gold label, to probe local decision boundaries and expose systematic gaps in standard test sets [9]. CheckList applies behavioral testing through capabilities, invariance tests, and directional expectation tests [26]. Metamorphic testing similarly specifies relations among multiple executions when a simple oracle is unavailable [3]. These approaches share an important principle: evidence about a model becomes more informative when its behavior is evaluated under controlled changes rather than on isolated, independently sampled inputs.

Recent software-engineering evaluations instantiate this principle directly. Ding et al. [7] use vulnerability-related paired evaluation to test whether code models distinguish closely related programs with different vulnerability labels. In satisfiability solving, Zhang et al. [42] introduce Accurate Differentiation Ratio (ADR) to measure whether a model correctly distinguishes matched SAT/UNSAT formulas. Such controlled metrics are closer to the intended semantic relation than marginal accuracy when the pairs are well constructed. They can suppress strategies that succeed only from class imbalance or cues shared within a pair.

Our work analyzes the remaining boundary of this approach. A controlled metric is still a metric over a particular intervention design. Label-changing edits may introduce residual artifacts, semantics-preserving transformations may fail to cover all nuisance factors, and repeated group or cluster success may remain correlated through shared shortcuts. We therefore distinguish the *design* of a semantic-changing or semantics-preserving intervention from the *metric* that aggregates its outcomes. ADR, group ADR, and cluster metrics provide stronger behavioral evidence than marginal prediction metrics, but the observed success can still combine semantic success with residual shortcut or guessing success. This motivates the non-separability analysis developed in this paper.

9.5 Explanation-Based and Mechanism-Oriented Evidence

Post-hoc explanation methods provide another source of evidence about why a model predicts as it does. For graph neural networks, GNNExplainer searches for a subgraph and feature subset that approximately preserve a model's prediction [40]. Integrated Gradients, SHAP, and DeepLIFT estimate feature or token contributions using gradients, perturbations, or reference activations [20, 30, 31]. Such methods can help diagnose whether a model appears to rely on vulnerability-relevant program structure, formula semantics, identifier names, API names, formatting, or other side-channel cues.

Explanation methods are complementary to prediction metrics because they retain or recover information that a scalar performance score discards. At the same time, an explanation must itself be validated: different explanation methods can allocate influence differently, and evidence that a feature influenced a prediction is not the same as evidence that the feature is semantically valid for the intended task. For this reason, explanations are most useful in our framework when they

motivate controlled interventions. If an attribution highlights a suspected side-channel factor, changing that factor while preserving task semantics provides a direct behavioral test of whether the prediction depends on it. Conversely, modifying the label-determining semantic relation while holding nuisance factors fixed tests semantic sensitivity.

9.6 Positioning of This Work

The preceding literature progressively enriches the evidence used to evaluate reasoning: outcome benchmarks measure final task success; trace metrics and process supervision inspect intermediate rationales; validity-centered work examines the connection between measurements and capability claims; and contrastive, metamorphic, and explanation-based methods probe model behavior under more informative designs. Our contribution is to characterize a common identification and validity boundary across these perspectives. We formalize why finite prediction and score observations do not, without additional assumptions, uniquely identify the mechanism responsible for benchmark success or guarantee that metric ordering is an ordering of reasoning ability. We further analyze why controlled pair, group, and cluster metrics reduce some shortcut opportunities without fully separating semantic success from residual non-semantic success.

Accordingly, the conclusion of this work is not that reasoning metrics are useless. Different metrics are valid for different observable criteria, and richer evaluation designs support stronger behavioral claims. The central question is instead how far those claims may be extended. We use checkable SAT witnesses, controlled interventions, and nuisance-factor audits to study how additional information narrows the gap between benchmark performance and reasoning claims. This places our work between metric analysis and construct-validity analysis: we treat metrics as useful summaries of observable behavior while making explicit the additional assumptions and evidence required to interpret those summaries as evidence of reasoning.

10 Conclusion

This paper distinguishes the observable criterion computed by an evaluation metric from the broader construct of reasoning ability. Metrics based on finite labels, predictions, or scalar scores cannot in general identify the mechanism that produced success. Their values need not be monotone in semantic reasoning, and controlled-success probabilities do not uniquely separate semantic from residual non-semantic success.

The empirical reanalysis shows that these are practical, not merely philosophical, boundaries. Conventional metrics remain nontrivial after constructive witness production collapses; model orderings reverse relative to witness-aware evidence; and controlled results change under pair rematching, effect thresholds, confidence requirements, and shortcut-ceiling specifications. Matched and confidence-bounded metrics are valuable precisely because they expose some of these failures. Group analysis adds a further boundary: no group reaches ten successful matched pairs in 88/130 model-size slices, even though isolated pair success is common. These metrics still measure behavior under a selected design rather than a design-independent latent mechanism.

The appropriate conclusion is therefore not that reasoning evaluation is impossible or that metrics are useless. It is that claim strength must track evidence strength. Accuracy and F1 support claims about benchmark agreement. ADR and matched separability support claims about controlled contrast behavior. Group thresholds support claims about persistence across controlled variants. Checkable witnesses support constructive correctness claims and are the closest machine-checkable proxy to demonstrated reasoning in the constraint archives. Perturbations and nuisance-factor audits support conditional invariance claims. Stronger claims about reasoning require these sources of evidence to be combined and their remaining assumptions made explicit.

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